

Precautionary Pricing and Markup Cyclicalit^{*}

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Abstract

Why are markups cyclical, and why does their cyclicalit^y vary across firms, sectors, and business-cycle episodes? We develop a theory in which risk-averse firms post prices before costs and demand are realized and use prices to manage their risk exposure. Cost uncertainty induces firms to raise prices to limit sales in high-cost states, whereas demand uncertainty induces them to lower prices to support sales in weak-demand states. These opposing responses imply that markup cyclicalit^y depends on the composition and cyclicalit^y of risk, as well as on realized marginal-cost shocks. We test these predictions using U.S. manufacturing data. A one-standard-deviation increase in cost uncertainty raises markups by 1.3 percent, while the same increase in demand uncertainty lowers them by 1.2 percent. Using these estimates, we decompose markup cyclicalit^y and find that realized marginal-cost shocks dominate overall, while uncertainty matters more in earlier business-cycle episodes and recessions. We embed the precautionary-pricing mechanism in a general-equilibrium model of firm dynamics. Eliminating precautionary pricing may appear welfare-improving because lower markups increase allocative efficiency. In general equilibrium, however, it also induces substantial firm exit and reduces product variety, offsetting the welfare gains from lower markups.

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1 Introduction

Why are markups cyclical, and why does their cyclicality differ across firms, sectors, and business-cycle episodes? Empirical findings are strikingly heterogeneous: in some contexts markups appear countercyclical, in others procyclical, or even close to acyclical. Despite broad agreement on the cyclical behavior of consumption, investment, and employment, there is no consensus on the cyclical behavior of markups.

This paper argues that part of this heterogeneity reflects differences in the risks firms face when setting prices. When a firm chooses its price before costs and demand are realized, the price determines not only expected profits but also how profits vary across states. A risk-averse firm therefore has an incentive to use its price to manage its exposure to uncertainty. The direction of this precautionary pricing response depends on the source of risk.

Consider first uncertainty about cost. A low posted price commits the firm to supplying more units when costs turn out to be high, making profits more exposed to adverse cost realizations. By raising its price, the firm reduces sales and limits this exposure. Cost uncertainty therefore induces the firm to raise its price and the target markup. Demand uncertainty works in the opposite direction. A larger price-cost wedge makes profits more sensitive to demand realizations. By lowering its price toward marginal cost, the firm sacrifices some profits in strong-demand states while supporting sales in weak-demand states. Demand uncertainty therefore induces the firm to compress its markup.

These opposing pricing responses provide a simple mechanism for markup cyclicity. When uncertainty varies over the business cycle, the cyclicity of the markup firms target depends both on the composition of risk and on how each source of risk co-moves with aggregate activity. Procyclical cost uncertainty pushes targeted markups toward procyclicality, whereas procyclical demand uncertainty pushes them toward countercyclicality. The same economy can therefore exhibit different markup cyclicity across firms, sectors, or episodes as the composition and cyclicity of risk change.

The markup measured in the data—after production occurs—contains an additional component. Firms set prices in response to the distribution of shocks they anticipate, while measured marginal cost also reflects the costs subsequently realized. Measured markup cyclicity therefore combines a forward-looking uncertainty channel, operating through ex-ante pricing decisions, with a realized-marginal-cost channel. At the aggregate level, reallocation across firms and sectors provides a further source of movement in measured markups.

We develop a general theory of this mechanism in which risk-averse firms post prices before uncertainty is resolved and subsequently satisfy demand at those prices. Optimal

pricing can be separated into the standard risk-neutral pricing and a precautionary adjustment reflecting the value of redistributing profits across states. States in which the entrepreneur's total resources are low receive greater weight because marginal utility is high. Around the risk-neutral benchmark, the precautionary price adjustment depends on the entrepreneur's risk aversion, cost pass-through, and the extent to which firm-level risk is ultimately borne by the entrepreneur. Non-firm resources and risk-sharing arrangements attenuate the mechanism, while complete insurance eliminates the corresponding precautionary response.

Main Contributions. Our first contribution is theoretical. We characterize precautionary pricing around the risk-neutral benchmark and derive source-specific predictions for different forms of uncertainty. Under standard exposure conditions, cost uncertainty raises prices and markups, whereas demand uncertainty lowers them. We then show how time variation in these risks maps into ex-ante markup cyclicalities and how realized marginal-cost shocks and reallocation shape the markup cyclicalities measured in the data. The framework thereby provides a unified interpretation of the heterogeneous empirical evidence on markup cyclicalities documented in the literature (e.g., [Bils, 1987](#); [Hall, 1988](#); [Basu and Fernald, 1997](#); [Nekarda and Ramey, 2020](#); [De Loecker et al., 2020](#); [Hasenzagl and Pérez, 2026](#)).

Our second contribution is empirical. We test the theory using industry-level data for U.S. manufacturing from 1972 to 2018. We measure markups using De Loecker–Warzynski cost-share markups, treating labor and materials as the flexible input bundle. We construct industry-specific, time-varying measures of cost and demand uncertainty from each industry's position in the production network. Cost uncertainty is the trailing volatility of innovations to a supplier-weighted input-price index, while demand uncertainty is the trailing volatility of surprises to a customer-weighted activity index. Both measures are constructed from the BEA input–output tables and exclude own-industry flows. To address measurement error, we instrument these uncertainty measures with leave-one-out shift-share (Bartik) instruments based on second moments, while controlling throughout for the corresponding first-moment cost and demand conditions. Consistent with the theory, a one-standard-deviation increase in cost uncertainty raises markups by +1.3 percent, whereas an equally sized increase in demand uncertainty lowers them by -1.2 percent. The same asymmetry appears within firms: a given firm raises its markup when cost uncertainty in its industry increases and lowers it when demand uncertainty rises. We then use these estimated responses to decompose markup cyclicalities into the contributions of cost uncertainty, demand uncertainty, and realized marginal-cost shocks. Realized marginal-cost shocks account for most cyclical variation overall, while uncertainty plays a larger role in earlier business-cycle episodes—before

the Great Moderation—and recessions. Within the uncertainty component, cost uncertainty contributes most strongly around the oil-shock episodes of the 1970s and early 1980s, whereas demand uncertainty plays a larger role around the dot-com recession and the Great Recession.

Our third contribution is quantitative. We embed the precautionary-pricing mechanism in a dynamic general-equilibrium model with heterogeneous firms, Melitz–Ottaviano preferences, free entry and exit, decreasing returns, and idiosyncratic and aggregate uncertainty in both cost and demand. Firms hold no assets and, conditional on operating, pricing is their only within-period instrument for managing exposure to firm-profit risk. We estimate the model by simulated method of moments, targeting the empirical responses of markups to cost and demand uncertainty together with the aggregate markup and the manufacturing firm exit rate. The model allows us to quantify the general-equilibrium consequences of precautionary pricing for markups, entry, product variety, and welfare. Eliminating precautionary pricing may appear welfare-improving because lower markups increase allocative efficiency. In general equilibrium, however, it also induces substantial firm exit and reduces product variety, offsetting the welfare gains from lower markups.

Related Literature. Our paper relates first to recent work showing that uninsurable risk can generate wedges between prices and marginal costs (Boar, Gorea, and Midrigan, 2025; Di Tella, Malgieri, and Tonetti, 2025). In these models, imperfectly diversified entrepreneurs choose inputs before productivity is realized, and risk aversion distorts these choices away from the risk-neutral benchmark. Our mechanism instead operates directly through the pricing decision. Firms post prices before cost and demand shocks are realized and subsequently satisfy demand, so a price change redistributes profits across states while quantities adjust ex post. We focus on this pricing margin because survey evidence suggests that price targeting is common among manufacturing firms (Aiginger, 1999), while recent work formalizes the choice between price and quantity targets and shows that price commitments can be optimal under plausible configurations of demand and cost uncertainty (Flynn, Nikolakoudis, and Sastry, 2026b). Our framework also accommodates multiple sources of uncertainty whose pricing effects differ in sign. Cost risk induces precautionary increases in markups, whereas demand risk induces precautionary reductions in markups. When demand risk is absent and uncertainty arises solely from productivity, our mechanism generates a risk component of markups analogous to that arising from precautionary input choice.

Contemporaneous work by Flynn, Nikolakoudis, and Sastry (2026a) also shows that uncertainty can affect prices and markups directly, but through a distinct mechanism. In their environment, firms choose both prices and production before demand and

productivity are realized, and realized sales equal the smaller of supply and demand. Product markets can therefore be slack or rationed. Uncertainty raises markups because production costs are incurred in every state while additional production generates revenue only when demand is rationed; firms respond by reducing production and raising prices. Their mechanism operates under expected-profit maximization and arises from joint price-production commitment and the failure of ex-post market clearing. In our environment, firms instead commit to a price but subsequently satisfy realized demand, eliminating the slack-versus-rationing margin. Uncertainty affects pricing because risk-averse entrepreneurs value profits differently across states. This distinction also generates different predictions across sources of uncertainty: in particular, demand uncertainty raises prices in their framework, whereas it lowers prices and markups through the precautionary-pricing mechanism in ours. The two papers therefore identify complementary channels through which uncertainty can shape firms' pricing decisions.

Our analysis also complements [Burstein, Carvalho, and Grassi \(2025\)](#). They develop a granular oligopolistic model in which firm-level shocks generate sectoral and aggregate markup dynamics through endogenous changes in market shares and demand elasticities. In their framework, shocks move firms along a markup–size schedule implied by variable demand curvature, generating within-firm markup changes as well as aggregate movements through reallocation. We provide a complementary structural mechanism for within-firm adjustments. In our framework, firms' markups change because uncertainty alters the precautionary value of the price they set, while reallocation provides an additional source of aggregate markup dynamics. Their mechanism and ours can therefore operate simultaneously.

More broadly, our paper contributes to the literature studying the cyclical behavior and aggregate consequences of markups. The traditional interpretation emphasizes market power and the response of markups to realized changes in demand, costs, and competitive conditions (e.g., [Chamberlin, 1933](#); [Dixit and Stiglitz, 1977](#); [Blanchard and Kiyotaki, 1987](#); [Edmond et al., 2023](#)). We complement this view by showing that even holding the underlying competitive environment fixed, changes in the distribution of future costs and demand can alter optimal markups before those shocks are realized. Precautionary pricing therefore provides a distinct source of within-firm markup variation that can coexist with market power and other pricing frictions.

Our mechanism also relates to the New Keynesian literature on forward-looking price setting under nominal rigidities (e.g., [Calvo, 1983](#); [Golosov and Lucas, 2007](#)). In those models, a firm's current price reflects the consequences of remaining at that price across future dates. Precautionary pricing introduces a different form of forward-looking behavior: because firms commit to prices before uncertainty is resolved, the

chosen price determines profits across future states. Risk aversion therefore alters the weights placed on those states even in the absence of nominal rigidities. More recent work emphasizes how firm-level demand conditions, demand curvature, and anticipated demand dispersion shape desired prices and markups in state-dependent pricing environments (Aruoba et al., 2024a,b). Our mechanism provides a complementary source of variation in desired markups through changes in the distribution of future cost and demand risk.

The remainder of the paper is organized as follows. Section 2 presents a simple example illustrating the core mechanism. Section 3 develops the general theory of pricing under risk and markup cyclicalities. Section 4 tests the theory’s predictions on U.S. manufacturing and provides a markup cyclicalities decomposition separating the contributions of uncertainty from those of realized shocks. Section 5 embeds the static theory in a dynamic general-equilibrium model with free entry, presents the results from structural estimation, and conducts counterfactual exercises. Section 6 concludes.

2 A Stylized Example

This section illustrates the precautionary-pricing mechanism in a simple environment. A firm sets its price before cost and demand shocks are realized and commits to satisfying demand at that price. When the firm’s owner is risk averse, price affects not only expected profit but also exposure to profit risk. As a result, cost risk induces the firm to raise its price and markup, whereas demand risk induces it to lower them.

The example also shows how these pricing responses translate into markup cyclicalities. The ex-ante markup reflects the firm’s response to the composition of risk when price is set, while the ex-post markup additionally reflects realized marginal-cost shocks. Measured markup cyclicalities therefore depend both on the risks firms face and on the shocks that are subsequently realized.

To isolate these mechanisms as transparently as possible, the example deliberately imposes strong assumptions: CARA preferences, normally distributed shocks, linear demand, constant marginal cost, no income outside the firm, and no other instruments for insuring firm-specific risk. These assumptions make pricing the entrepreneur’s only margin for managing risk. Section 3 relaxes these functional-form assumptions and insurance restrictions and makes explicit the conditions under which precautionary pricing remains operative.

Environment. A monopolistic firm sets its price p before demand and cost shocks are realized.¹ Demand is linear, $y(p; \xi) = a + \xi - bp$, where $a, b > 0$, and ξ is a demand shock. Marginal cost is constant in output but stochastic, $\kappa(\eta) = \bar{\kappa} + \eta$, where $\bar{\kappa} > 0$ and η is a cost shock. The firm commits to satisfying realized demand at its posted price, so profits are $\pi(p; \eta, \xi) = (p - \bar{\kappa} - \eta)(a + \xi - bp)$. The shocks η and ξ are independent, mean zero, and normally distributed, with variances σ_η^2 and σ_ξ^2 . The firm is owned by a risk-averse entrepreneur with CARA preferences, $U(\pi) = -\exp(-\alpha\pi)$, where $\alpha \geq 0$ is the entrepreneur's coefficient of absolute risk aversion. When $\alpha = 0$, the entrepreneur is risk neutral; larger values of α imply greater aversion to fluctuations in firm profits.

The entrepreneur chooses the posted price p^* before uncertainty is resolved to maximize expected utility,

$$p^* \in \arg \max_{p > 0} \mathbb{E} \left[U(\pi(p; \eta, \xi)) \right].$$

The entrepreneur receives no income outside the firm and cannot insure its idiosyncratic risks. Price is therefore the only choice variable through which the entrepreneur can alter exposure to profit risk.

When either source of uncertainty is considered in isolation, profits are normal, so the CARA-normal structure yields an exact certainty-equivalent representation of the pricing problem:

$$p^* \in \arg \max_{p > 0} \left\{ \mathbb{E}[\pi(p)] - \frac{\alpha}{2} \text{Var}(\pi(p)) \right\}.$$

We use these exact one-risk cases to build intuition. When both sources of uncertainty are present, profits also contain the interaction term $-\eta\xi$ and are no longer normally distributed. We therefore characterize their joint effect using a first-order approximation around the risk-neutral benchmark, which isolates the opposing contributions of cost and demand risk. We first characterize the risk-neutral benchmark ($\alpha = 0$), and then show how risk aversion distorts pricing away from this benchmark.

Risk-neutral benchmark. When $\alpha = 0$, the entrepreneur is risk neutral and the pricing problem reduces to maximizing expected profits $\mathbb{E}[\pi(p)] = (p - \bar{\kappa})(a - bp)$. The optimal risk-neutral price satisfies

$$p_{\text{RN}} = \frac{a + b\bar{\kappa}}{2b}.$$

¹Appendix A provides a directed-search microfoundation for price commitment in which demand uncertainty arises endogenously from buyer arrival risk. Cost uncertainty is standard and reflects risk in input prices or own productivity risk.

Although the firm's choice variable is price, it is useful for graphical purposes to express the same optimality condition in output space. Using inverse demand $p(y) = (a - y)/b$, the condition becomes

$$\underbrace{\frac{a - 2y}{b}}_{\text{mean marginal revenue}} = \underbrace{\bar{\kappa}}_{\text{mean marginal cost}}$$

Figure 1 illustrates the benchmark. The optimal price p_{RN} implies an expected quantity y_{RN} at which mean marginal revenue equals mean marginal cost.

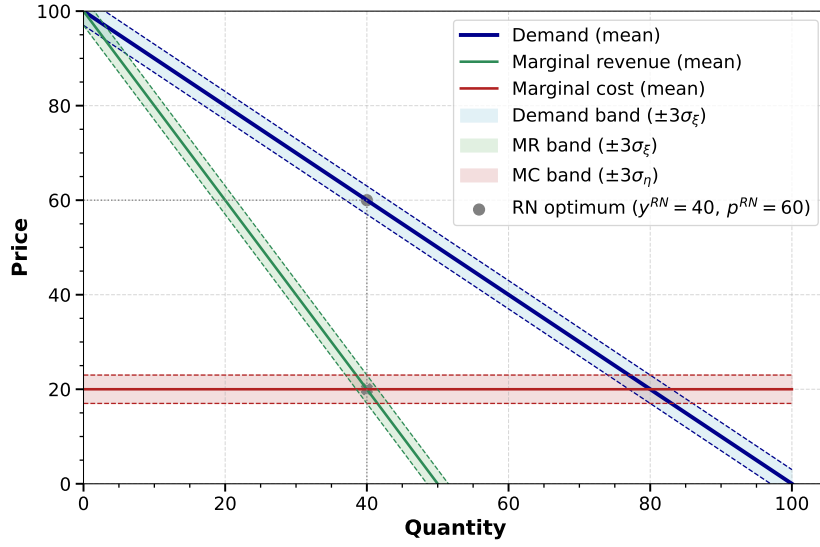


Figure 1: Optimal risk-neutral pricing.

Notes. Linear demand is $y = a - bp$ with $a = 100$ and $b = 1$. Inverse demand is $p(y) = 100 - y$, marginal revenue is $MR(y) = 100 - 2y$, and mean marginal cost is $\bar{\kappa} = 20$. Both bands are plotted with $\pm 3\sigma_i$ from their mean value, with $\sigma_i = 1$ for $i \in \{\eta, \xi\}$.

At p_{RN} , expected marginal profits are zero. A marginal change in price therefore has no first-order effect on expected profits. Locally around this benchmark, a price adjustment induced by risk aversion is instead governed by how changing price redistributes profits across states of the world.

Cost risk. Suppose first that only marginal cost is uncertain, so $\xi = 0$. Profits are $\pi(p; \eta) = (p - \bar{\kappa} - \eta)(a - bp)$, and profit variance is $\text{Var}(\pi(p)) = (a - bp)^2 \sigma_\eta^2 = y(p)^2 \sigma_\eta^2$. Exposure to cost risk therefore increases with the quantity implied by the posted price. If the firm sets a low price and thereby commits itself to supplying a large quantity, an adverse cost realization has a larger effect on profits. The firm's certainty equivalent is

$$\text{CE}(p) = (p - \bar{\kappa})(a - bp) - \frac{\alpha}{2}(a - bp)^2 \sigma_\eta^2 = y(p)^2 \sigma_\eta^2.$$

The optimality condition is

$$\frac{d}{dp}CE(p) = (a - 2bp + b\bar{\kappa}) + \alpha b(a - bp)\sigma_{\eta}^2 = 0.$$

For graphical purposes, this condition can again be represented in output space as

$$\underbrace{\frac{a - 2y}{b}}_{\text{mean marginal revenue}} = \underbrace{\bar{\kappa} + \alpha y \sigma_{\eta}^2}_{\text{risk-adjusted marginal cost}}$$

Cost uncertainty therefore acts like an endogenous, quantity-dependent adjustment to marginal cost. A posted price that implies greater sales exposes the entrepreneur more strongly to adverse cost realizations, so risk aversion effectively rotates the marginal-cost schedule upward. Figure 2 illustrates the resulting change in the optimal price.

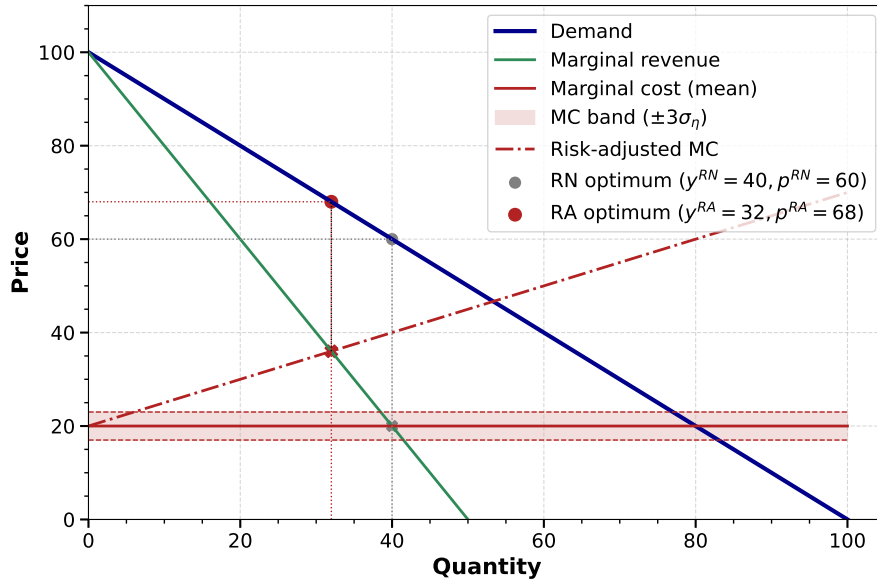


Figure 2: Optimal pricing with cost risk.

Notes. Linear demand is $y = a - bp$, with $a = 100$, $b = 1$, and $\bar{\kappa} = 20$. The marginal-cost band is plotted with bands $\bar{\kappa} \pm 3\sigma_{\eta}$.

A first-order expansion around the risk-neutral benchmark gives

$$p_{RA} \approx p_{RN} + \frac{\alpha}{2} y_{RN} \sigma_{\eta}^2,$$

so that

$$\frac{\partial p_{RA}}{\partial \sigma_{\eta}^2} > 0.$$

*Cost risk raises prices and markups.*² By raising its posted price, the firm reduces expected demand and therefore the quantity it expects to supply in high-cost states. The price increase sacrifices expected profits at the margin but provides partial insurance against adverse cost shocks. Because expected marginal cost remains fixed at $\bar{\kappa}$, the higher price also raises the markup targeted when the price is set, $\mu^{\text{ea}} = p / \bar{\kappa}$. The firm thus uses its price—and consequently its markup—to limit its exposure to cost risk.

Demand risk. Suppose instead that only demand is uncertain, so $\eta = 0$. Profits are $\pi(p; \xi) = (p - \bar{\kappa})(a + \xi - bp)$, and profit variance is $\text{Var}(\pi(p)) = (p - \bar{\kappa})^2 \sigma_\xi^2$. Exposure to demand risk therefore increases with the price-cost wedge. A larger markup makes profits more sensitive to whether demand turns out to be high or low. The optimality condition in this case is

$$\frac{d}{dp} \text{CE}(p) = (a - 2bp + b\bar{\kappa}) - \alpha(p - \bar{\kappa})\sigma_\xi^2 = 0.$$

Represented in output space, this condition becomes

$$\underbrace{\frac{a - 2y}{b} + \frac{\alpha}{b}(p(y) - \bar{\kappa})\sigma_\xi^2}_{\text{risk-adjusted marginal revenue}} = \underbrace{\bar{\kappa}}_{\text{mean marginal cost}}.$$

Demand uncertainty therefore changes the firm's effective marginal return to posting a price that implies lower sales. Because the price-cost wedge determines how strongly profits respond to demand realizations, the entrepreneur can reduce exposure to demand risk by compressing that wedge. Figure 3 shows that the risk-averse firm consequently chooses a lower price than the risk-neutral firm, which implies higher expected output.

A first-order expansion around the risk-neutral benchmark gives

$$p_{\text{RA}} \approx p_{\text{RN}} - \frac{\alpha}{2b}(p_{\text{RN}} - \bar{\kappa})\sigma_\xi^2,$$

so that

$$\frac{\partial p_{\text{RA}}}{\partial \sigma_\xi^2} < 0.$$

Demand risk lowers prices and markups. A high price places a larger price-cost wedge on every unit sold and thereby makes profits more sensitive to demand realizations.

²This mechanism echoes [Knight \(1921\)](#)'s view that bearing uninsurable risk can command compensation. In our setting, cost risk generates a precautionary component of the firm's markup relative to the otherwise identical risk-neutral benchmark.

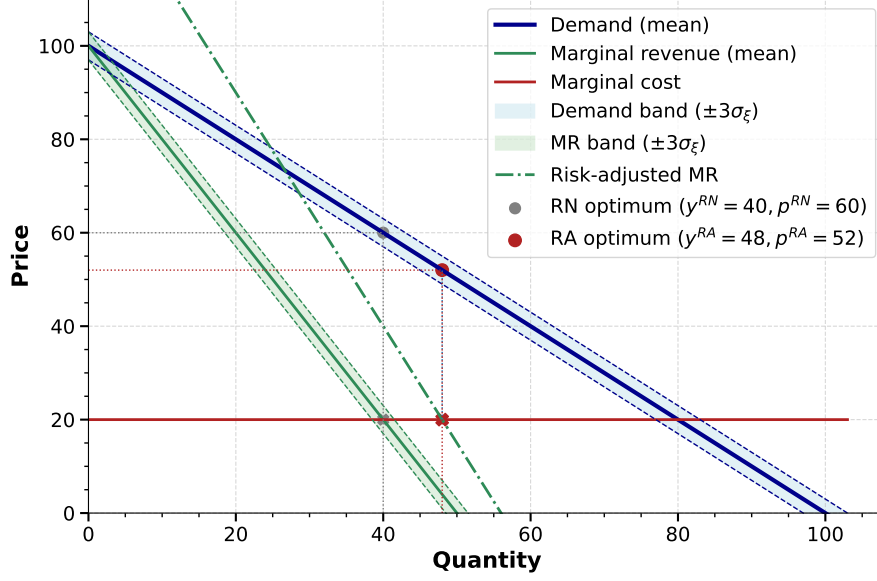


Figure 3: Optimal pricing with demand risk.

Notes. Linear demand is $y = a - bp$, with $a = 100$, $b = 1$, and $\bar{\kappa} = 20$. Demand- and marginal-revenue bands are plotted with $\pm 3\sigma_\xi$, where $\sigma_\xi^2 = 1$.

By lowering its price toward marginal cost, the firm sacrifices some profits in strong-demand states while supporting sales in weak-demand states. The firm thus compresses its markup to limit exposure to demand risk.

Markup cyclicality. Cost and demand risk generate opposing precautionary pricing incentives. Combining their first-order effects around the risk-neutral benchmark gives

$$p_{RA,t} \approx p_{RN,t} + \alpha \mathcal{R}(\sigma_t^2),$$

where

$$\mathcal{R}(\sigma_t^2) = \frac{y_{RN}}{2} \left(\sigma_{\eta,t}^2 - \frac{1}{b^2} \sigma_{\xi,t}^2 \right).$$

This expression abstracts from higher-order interactions between the two shocks. It shows that the precautionary price adjustment depends on the *composition of risk*, rather than simply on the total amount of uncertainty: cost risk raises the optimal price, whereas demand risk lowers it.

To connect these pricing effects to markups, it is useful to distinguish between ex-ante and ex-post notions of the markup. Define the *ex-ante markup*—the one targeted

by the firm when setting price and before shocks are realized—as

$$\mu_t^{\text{ea}} := \frac{p_t}{\mathbb{E}[\kappa_t]} = \frac{p_t}{\bar{\kappa}}.$$

The pricing result above implies

$$\mu_t^{\text{ea}} \approx \mu_{\text{RN}} + \frac{\alpha}{\bar{\kappa}} \mathcal{R}(\sigma_t^2).$$

Variation in the composition of risk therefore translates directly into variation in the markup targeted by the firm. Let Y_t denote aggregate economic activity. Using the first-order pricing approximation above,

$$\text{Cov}(\mu_t^{\text{ea}}, Y_t) \approx \frac{\alpha}{\bar{\kappa}} \left[\frac{\partial \mathcal{R}}{\partial \sigma_\eta^2} \text{Cov}(\sigma_{\eta,t}^2, Y_t) + \frac{\partial \mathcal{R}}{\partial \sigma_\xi^2} \text{Cov}(\sigma_{\xi,t}^2, Y_t) \right].$$

Each source of risk therefore contributes to ex-ante markup cyclicalities through two objects: how strongly the firm's markup responds to that risk, as indicated by $\partial \mathcal{R} / \partial \sigma_\bullet^2$, and how the risk itself co-moves with economic activity, which is captured by $\text{Cov}(\sigma_\bullet^2, Y_t)$. Since cost risk raises prices ($\partial \mathcal{R} / \partial \sigma_\eta^2 > 0$) and demand risk lowers them ($\partial \mathcal{R} / \partial \sigma_\xi^2 < 0$), countercyclical cost uncertainty makes ex-ante markups more countercyclical and demand uncertainty makes them more procyclical. If the two forces offset, ex-ante markups can appear approximately acyclical even when firms respond strongly to uncertainty. The cyclicalities of the markup thus depend on the composition of risk.

The *ex-post markup* instead uses realized marginal cost. Once the cost shock η_t is realized, marginal cost is $\bar{\kappa} + \eta_t$, so³

$$\mu_t^{\text{ep}} := \frac{p_t}{\kappa_t} = \frac{p_t}{\bar{\kappa} + \eta_t} = \frac{\bar{\kappa}}{\bar{\kappa} + \eta_t} \mu_t^{\text{ea}}.$$

The distinction between ex-ante and ex-post markups is important because the markup we estimate in the data is an ex-post object. It therefore reflects both the firm's pricing response to the distribution of shocks it anticipates when setting its price and the realization of marginal cost after the price has been chosen.

Unlike the preceding characterization of ex-ante markup cyclicalities, the relationship between ex-ante and ex-post markups is exact. We can write

$$\mu_t^{\text{ep}} = \mu_t^{\text{ea}} - \frac{\eta_t}{\bar{\kappa} + \eta_t} \mu_t^{\text{ea}}.$$

³For ease of exposition, marginal cost is constant in output in the current example. More generally, realized demand shocks also affect the ex-post markup when marginal cost varies with output. In that case, demand realizations change the scale of production and thereby realized marginal cost.

Taking covariances with aggregate activity therefore yields the exact decomposition

$$\text{Cov}(\mu_t^{\text{ep}}, Y_t) = \underbrace{\text{Cov}(\mu_t^{\text{ea}}, Y_t)}_{\text{uncertainty channel}} - \underbrace{\text{Cov}\left(\frac{\eta_t}{\bar{\kappa} + \eta_t} \mu_t^{\text{ea}}, Y_t\right)}_{\text{realized-cost channel}}.$$

This decomposition separates the two forces that determine measured markup cyclical-ity in the example. The first is forward looking: firms adjust prices to the composition of the risks they face before shocks are realized. The second operates through realized marginal costs after prices have been set. Measured markup cyclical-ity therefore re-flects both the composition of risk and the interaction between realized shocks and the ex-ante markup chosen before those shocks are observed.

The example isolates the paper’s central mechanism in a deliberately stark environ-ment. The next section generalizes these insights, characterizing the conditions under which precautionary pricing arises with general preferences, demand and cost func-tions, and incomplete insurance, and deriving their implications for markup cyclical-ity. The remainder of the paper then asks whether these predictions are borne out in the data and whether the mechanism is quantitatively important in general equilibrium.

3 A General Theory of Markup Cyclical-ity

We consider a firm that posts its price before current demand and cost shocks are realized. The firm is owned by an entrepreneur who chooses price to maximize expected utility. Unlike the stylized example, demand and technology need not take particular functional forms, marginal cost may vary with output, the entrepreneur may receive more income outside the firm and may have access to risk-sharing arrangements that partially or fully insure some shocks. We characterize how these features shape precautionary pricing and, in turn, markup cyclical-ity at various levels of aggregation.

3.1 General pricing under uncertainty

Let θ denote a vector of demand and cost shocks. Demand for the firm’s variety is $y(p; \theta) = \mathcal{D}(p; \theta)$, total cost is $\mathcal{C}(y; \theta)$, and profits are

$$\pi(p; \theta) = p\mathcal{D}(p; \theta) - \mathcal{C}(\mathcal{D}(p; \theta); \theta).$$

Let $\omega(\theta)$ denote the entrepreneur's resources outside the firm. These resources may be stochastic and correlated with the shocks affecting firm profits. Total resources are

$$\mathcal{W}(p; \theta) = \omega(\theta) + \pi(p; \theta),$$

and the entrepreneur chooses p to solve

$$\max_{p > 0} \mathbb{E}_{\theta} [U(\mathcal{W}(p; \theta))].$$

Define realized marginal cost at the quantity induced by p as

$$\mathcal{MC}(p; \theta) := \mathcal{C}_y(\mathcal{D}(p; \theta); \theta).$$

Theorem 1 (Optimal pricing under uncertainty). *Suppose that:*

- A1. **Price commitment.** *The firm sets p before θ is realized and commits to satisfying demand at that price, $y = \mathcal{D}(p; \theta)$. Thus output is determined by the posted price and realized demand, and is not subsequently adjusted in response to shocks.*
- A2. **Preferences.** *U is twice continuously differentiable, strictly increasing, and strictly concave.*
- A3. **Demand and technology.** *For almost all θ , $\mathcal{D}(\cdot; \theta)$ and $\mathcal{C}(\cdot; \theta)$ are twice continuously differentiable, with $\mathcal{D}_p(p; \theta) < 0$ and $\mathcal{C}_y(y; \theta) > 0$.*
- A4. **Interior optimum.** *The expected-utility problem admits a unique interior optimum.*
- A5. **Regularity.** *The expected-utility objective is finite and continuously differentiable in p in a neighborhood of the optimum, with its derivative obtained by differentiation under the expectation.*

Then the unique optimal price p^* satisfies

$$\mathbb{E}_{\theta} [U'(\mathcal{W}(p^*; \theta)) \pi_p(p^*; \theta)] = 0,$$

where

$$\pi_p(p; \theta) = y(p; \theta) + [p - \mathcal{MC}(p; \theta)] y_p(p; \theta).$$

Proof. See Appendix B.1. □

Theorem 1 makes explicit how risk aversion changes the standard monopoly-pricing condition. Marginal profits are weighted by the entrepreneur's marginal utility in each

state. States in which total resources are low therefore receive greater weight than under risk neutrality. Because price changes both revenue and the quantity the firm supplies, the firm can use its posted price to alter its exposure to these states.

The first-order condition also admits a useful marginal-revenue interpretation. Define the risk-adjusted expectation at price p by

$$\mathbb{E}_p^{\text{RA}}[X] := \frac{\mathbb{E}[U'(\mathcal{W}(p; \theta))X]}{\mathbb{E}[U'(\mathcal{W}(p; \theta))]}.$$

Corollary 1 (Risk-adjusted pricing formula). *The optimal price satisfies*

$$\mathcal{MR}_{\text{RA}}(p^*) = \mathcal{MC}_{\text{RA}}(p^*),$$

where risk-adjusted marginal revenue is

$$\mathcal{MR}_{\text{RA}}(p) := p + \frac{\mathbb{E}_p^{\text{RA}}[y(p; \theta)]}{\mathbb{E}_p^{\text{RA}}[y_p(p; \theta)]} \equiv \underbrace{p + \frac{\mathbb{E}[y(p; \theta)]}{\mathbb{E}[y_p(p; \theta)]}}_{\mathcal{MR}_{\text{mean}}(p)} + \underbrace{\left(\frac{\mathbb{E}_p^{\text{RA}}[y]}{\mathbb{E}_p^{\text{RA}}[y_p]} - \frac{\mathbb{E}[y]}{\mathbb{E}[y_p]} \right)}_{\mathcal{RA}_{\text{MR}}(p)}$$

and risk-adjusted marginal cost is

$$\mathcal{MC}_{\text{RA}}(p) := \frac{\mathbb{E}_p^{\text{RA}}[\mathcal{MC}(p; \theta)y_p(p; \theta)]}{\mathbb{E}_p^{\text{RA}}[y_p(p; \theta)]} \equiv \underbrace{\frac{\mathbb{E}[\mathcal{MC}(p; \theta)y_p(p; \theta)]}{\mathbb{E}[y_p(p; \theta)]}}_{\mathcal{MC}_{\text{mean}}(p)} + \underbrace{\left(\frac{\mathbb{E}_p^{\text{RA}}[\mathcal{MC} y_p]}{\mathbb{E}_p^{\text{RA}}[y_p]} - \frac{\mathbb{E}[\mathcal{MC} y_p]}{\mathbb{E}[y_p]} \right)}_{\mathcal{RA}_{\text{MC}}(p)}.$$

Proof. See Appendix B.7. □

Optimal pricing equates risk-adjusted marginal revenue and risk-adjusted marginal cost. Relative to the risk-neutral benchmark, risk aversion tilts the effective probability distribution toward states in which marginal utility is high—that is, states in which the entrepreneur’s total resources are low. This reweighting modifies both marginal revenue and marginal cost by assigning greater weight to states in which the entrepreneur is exposed to more risk. Prices therefore serve as an instrument for managing exposure to risk. By adjusting prices, firms trade off expected profits against lower exposure to states in which marginal utility is high. When utility is linear, the risk-adjusted expectations collapse to physical expectations and the standard risk-neutral pricing rule is recovered.

3.2 Precautionary pricing around the risk-neutral benchmark

The exact characterizations in Theorem 1 and Corollary 1 are fully general, but they do not reveal how particular sources of uncertainty affect prices. To obtain sharper economic insight, we now study pricing locally around the risk-neutral benchmark. This approximation isolates the precautionary component of pricing and provides intuition for how the distribution of shocks affects optimal prices. Before deriving the local pricing formula, we introduce cost pass-through, which will be useful for expressing the local pricing results.

Definition 1 (Cost pass-through). *Let $\bar{\kappa}$ be a deterministic cost shifter entering the profit function $\pi(p; \theta, \bar{\kappa})$. Let the risk-neutral price $p_{RN}(\bar{\kappa})$ solve*

$$\mathbb{E}_\theta[\pi_p(p_{RN}(\bar{\kappa}); \theta, \bar{\kappa})] = 0.$$

The risk-neutral cost pass-through is defined as

$$\varphi := \frac{\partial p_{RN}(\bar{\kappa})}{\partial \bar{\kappa}}.$$

Differentiating the risk-neutral first-order condition and applying the implicit function theorem yields

$$\varphi = -\frac{\mathbb{E}_\theta[\pi_{p\bar{\kappa}, RN}]}{\mathbb{E}_\theta[\pi_{pp, RN}]},$$

where the subscript RN denotes evaluation at the risk-neutral optimum. This expression does not require constant marginal cost. If $\bar{\kappa}$ enters marginal cost additively and one-for-one, as with a unit tax or an additive input-price shock, so that

$$\mathcal{MC}(y; \theta, \bar{\kappa}) = \widetilde{\mathcal{MC}}(y; \theta) + \bar{\kappa},$$

then $\pi_{p\bar{\kappa}, RN} = -y_{p, RN}$, and therefore

$$\varphi = \frac{\mathbb{E}_\theta[y_{p, RN}]}{\mathbb{E}_\theta[\pi_{pp, RN}]}.$$

Since $\mathbb{E}_\theta[y_{p, RN}] < 0$ and $\mathbb{E}_\theta[\pi_{pp, RN}] < 0$, we have $\varphi > 0$. Thus, conditional on the local demand slope, pass-through summarizes the curvature cost of changing prices in the risk-neutral problem: larger φ corresponds to a locally flatter objective, allowing the firm to move prices more at lower expected cost to manage profit risk through pricing.

Theorem 2 (Local pricing formula around the risk-neutral price). *Suppose A1–A5 of Theorem 1 hold. Let the risk-neutral price p_{RN} be the unique interior maximizer of expected*

profits, characterized by

$$\mathbb{E}_\theta[\pi_{p,RN}] = 0, \quad \mathbb{E}_\theta[\pi_{pp,RN}] < 0.$$

In addition, suppose:

A6. Local preference family. Preferences belong to a one-parameter family $\{U_\alpha\}_{\alpha \geq 0}$ that nests risk neutrality at $\alpha = 0$, normalized so that $U'_0(\mathcal{W}) = 1$, and whose coefficient of absolute risk aversion satisfies

$$A_\alpha(\mathcal{W}) := -\frac{U''(\mathcal{W})}{U'(\mathcal{W})} = \alpha A(\mathcal{W}),$$

where $A(\mathcal{W}) > 0$ is continuously differentiable on the relevant domain.

A7. Regularity. The expected-utility objective is twice continuously differentiable in (p, α) in a neighborhood of $(p_{RN}, 0)$, with its derivatives obtained by differentiation under the expectation.

Define the benchmark objects evaluated at p_{RN} as

$$\begin{aligned} \mathcal{W}_{RN}(\boldsymbol{\theta}) &:= \omega(\boldsymbol{\theta}) + \pi(p_{RN}; \boldsymbol{\theta}), & \pi_{RN}(\boldsymbol{\theta}) &:= \pi(p_{RN}; \boldsymbol{\theta}), \\ \pi_{p,RN}(\boldsymbol{\theta}) &:= \pi_p(p_{RN}; \boldsymbol{\theta}), & y_{p,RN}(\boldsymbol{\theta}) &:= y_p(p_{RN}; \boldsymbol{\theta}). \end{aligned}$$

For any arbitrary fixed reference level $\mathcal{W}_*$, define

$$G(\mathcal{W}) := \int_{\mathcal{W}_*}^{\mathcal{W}} A(s) ds,$$

and let $\bar{A} := \mathbb{E}_\theta[A(\mathcal{W}_{RN})]$.

Then, for sufficiently small α , the optimal price $p(\alpha)$ satisfies

$$p(\alpha) = p_{RN} + \alpha \mathcal{R} + o(\alpha),$$

where the first-order risk adjustment admits the covariance representation

$$\mathcal{R} = \varphi \left[\underbrace{\frac{\bar{A} \cdot \text{Cov}_\theta(\mathcal{W}_{RN}, \pi_{p,RN})}{\mathbb{E}_\theta[y_{p,RN}]}}_{\text{pure-risk motive}} + \underbrace{\frac{\text{Cov}_\theta(G(\mathcal{W}_{RN}) - \bar{A}\mathcal{W}_{RN}, \pi_{p,RN})}{\mathbb{E}_\theta[y_{p,RN}]}}_{\text{wealth effects}} \right].$$

Proof. See Appendix B.2. □

Theorem 2 provides a transparent decomposition of precautionary pricing. Beyond the entrepreneur's degree of risk aversion, α , the magnitude of the precautionary price adjustment is linked to pass-through, φ , which summarizes how strongly optimal prices respond to changes in costs. Higher pass-through corresponds to a pricing problem in which larger price adjustments entail a smaller expected-profit cost.

The first term captures the pure-risk motive. Because a price change affects profits differently across states, the entrepreneur can use pricing to redistribute resources across realizations of uncertainty. If raising the price increases marginal profits primarily in states where total resources are low, then a higher price provides insurance and the entrepreneur sets a price above the risk-neutral benchmark. If raising the price instead increases marginal profits primarily in states where total resources are already high, then a higher price worsens risk exposure and the entrepreneur lowers the price relative to the risk-neutral benchmark. When non-profit resources are constant across states, total-resource risk coincides with profit risk, and this mechanism has a simple profit-smoothing interpretation.

The second term captures wealth effects. The pure-risk motive evaluates these transfers using average risk aversion, whereas the wealth-effects term recognizes that the value of insurance may itself vary across states. If the entrepreneur is more risk averse in the states that benefit from a price adjustment, wealth effects reinforce the pure-risk motive; if the entrepreneur is less risk averse in those states, they dampen it and can, in principle, overturn the overall pricing response. Under CARA preferences, absolute risk aversion is constant and this second channel disappears. Because non-profit resources, $\omega(\theta)$, may be arbitrarily correlated with profits, they can insure firm-level shocks and, when risk aversion varies with wealth, also affect the strength of the wealth-effects channel.

Corollary 2 (CARA and CRRA risk corrections). *Under the assumptions of Theorem 2:*

- (i) **CARA.** *Consider CARA preferences, represented for $\alpha > 0$ by $U_\alpha(W) = -\exp(-\alpha W)$. Under the normalization in A6, absolute risk aversion is constant, $A_\alpha(W) = \alpha$. The wealth-effects term therefore vanishes, and the risk adjustment simplifies to*

$$\mathcal{R} = \varphi \cdot \frac{\text{Cov}_\theta(W_{RN}, \pi_{p,RN})}{\mathbb{E}_\theta[y_{p,RN}]}.$$

If non-profit resources are constant across states, this further reduces to

$$\mathcal{R} = \varphi \cdot \frac{\text{Cov}_\theta(\pi_{RN}, \pi_{p,RN})}{\mathbb{E}_\theta[y_{p,RN}]}.$$

- (ii) **CRRA.** Consider CRRA preferences $U_\alpha(W) = W^{1-\alpha}/(1-\alpha)$, with the logarithmic specification at $\alpha = 1$. Then $A_\alpha(W) = \alpha/W$, and the general decomposition applies with a non-vanishing wealth-effects term.

To isolate the core precautionary pricing mechanism and its implications for markup cyclicity, we therefore focus first on the pure-risk component of the risk adjustment. The local pricing formula shows that this component depends on how marginal price changes redistribute total resources across states. Decomposing this exposure allows us to identify how distinct sources of uncertainty—cost risk, demand-level risk, and demand-elasticity risk—shape pricing incentives. Focusing on the pure-risk term isolates the precautionary role of pricing from wealth effects, which arise when absolute risk aversion varies with total resources.

Proposition 1 (Decomposition of the pure-risk motive by source). *Let $\theta = (\eta, \xi, \psi)$ collect a marginal-cost shifter η , a demand-level shifter ξ , and a demand-elasticity shifter ψ . Suppose the assumptions of Theorem 2 hold. In addition, assume:*

*A8. **Shock structure.** The shocks have mean zero, are mutually independent, and have finite fourth moments.*

*A9. **Local smoothness.** For each p in a neighborhood of p_{RN} , the mappings $\theta \mapsto \pi(p; \theta)$ and $\theta \mapsto \omega(\theta)$ are sufficiently smooth in a neighborhood of $\theta = \mathbf{0}$.*

Let $\sigma_k^2 = \text{Var}(\theta_k)$, $\Sigma = \text{Var}(\theta)$, and, for $k, l \in \{\eta, \xi, \psi\}$, define the local exposures

$$\mathcal{W}_k := \left. \frac{\partial \mathcal{W}(p; \theta)}{\partial \theta_k} \right|_{(p, \theta) = (p_{RN}, \mathbf{0})} \quad \text{and} \quad \pi_{pk} := \left. \frac{\partial \pi_p(p; \theta)}{\partial \theta_k} \right|_{(p, \theta) = (p_{RN}, \mathbf{0})}.$$

Let $\mathcal{R}$ denote the risk adjustment from Theorem 2, and write

$$\mathcal{R} = \mathcal{R}_{risk} + \mathcal{R}_{wealth},$$

where $\mathcal{R}_{risk}$ denotes the pure-risk component and $\mathcal{R}_{wealth}$ the wealth-effects component.

Then the pure-risk component decomposes as

$$\mathcal{R}_{risk} = \sum_{k \in \{\eta, \xi, \psi\}} \mathcal{R}_{risk}^{(k)} + o(\|\Sigma\|),$$

where

$$\mathcal{R}_{risk}^{(k)} := \varphi \bar{A} \cdot \frac{\mathcal{W}_k \pi_{pk}}{\mathbb{E}_\theta[y_{p, RN}]} \sigma_k^2.$$

Proof. See Appendix B.3. □

Proposition 1 shows that each source of uncertainty affects precautionary pricing through two exposures. The term $\mathcal{W}_k$ measures how shock k affects the entrepreneur's total resources, while π_{pk} measures how the marginal effect of a price increase on profits varies with the shock. Together, they determine whether changing the price offsets or reinforces the effect of shock k on the entrepreneur's resources. If a marginal price increase improves profits relatively more in states in which total resources are low, the entrepreneur raises the ex-ante price; if it improves profits relatively more in states in which total resources are high, the entrepreneur lowers it.

The distinction between total resources and firm profits is important because the entrepreneur need not bear the firm's shocks one-for-one. Non-profit income such as asset holdings or explicit hedging positions may offset part of the exposure generated by the firm. For example, airlines may hedge fuel-price risk, commodity-intensive producers may hedge input costs using futures or forward contracts, and exporters may hedge exchange-rate risk. Such arrangements are common when the underlying risk is readily tradable, especially for commodities and currencies, but they are neither universal nor typically complete. In the theory, these forms of insurance reduce the extent to which a shock to firm profits translates into a shock to the entrepreneur's total resources. Partial insurance therefore weakens the incentive to use prices to manage that risk, while complete insurance eliminates it.

In the absence of such insurance, $\omega_k = 0$ and therefore $\mathcal{W}_k = \pi_k$. The source-specific pricing response then depends on two simple questions: whether shock k raises or lowers profits, and whether it raises or lowers the marginal benefit of increasing the price. The next proposition uses these exposures to derive sufficient conditions under which cost, demand-level, and demand-elasticity risk raise or lower prices.

Proposition 2 (Sufficient conditions to sign each source of risk). *Suppose the assumptions of Proposition 1 hold. Assume that, locally at $(p, \theta) = (p_{RN}, \mathbf{0})$, the firm-side exposures satisfy:*

$$\pi_\eta < 0, \quad \pi_{p\eta} > 0; \quad \pi_\xi > 0, \quad \pi_{p\xi} \geq 0; \quad \pi_\psi < 0, \quad \pi_{p\psi} < 0;$$

and that the corresponding total-resource exposures preserve the sign of the firm-profit exposures,

$$\mathcal{W}_\eta < 0, \quad \mathcal{W}_\xi > 0, \quad \mathcal{W}_\psi < 0.$$

Then:

1. *Cost uncertainty raises prices: $\mathcal{R}_{risk}^{(\eta)} > 0$.*
2. *Demand-level uncertainty lowers prices: $\mathcal{R}_{risk}^{(\xi)} \leq 0$, with strict inequality if $\pi_{p\xi} > 0$.*

3. Demand-elasticity uncertainty lowers prices: $\mathcal{R}_{risk}^{(\psi)} < 0$.

Proof. See Appendix B.4. □

The exposure conditions in Proposition 2 have a natural economic interpretation. Higher marginal costs reduce firm profits but raise the marginal value of increasing prices; stronger demand raises both profits and the marginal payoff from higher prices; and greater demand elasticity lowers both. These conditions are satisfied in standard monopolistic-pricing environments and describe how price changes redistribute profits across states. The additional sign restrictions on total resources allow for partial insurance through non-profit income, but require that such insurance not fully eliminate or reverse the entrepreneur's exposure to the underlying shock. Full insurance, corresponding to $\mathcal{W}_k = 0$, eliminates the corresponding pure-risk pricing motive, while sufficiently strong offsetting exposure can reverse its sign.

The proposition therefore characterizes the pricing response in terms of the entrepreneur's *net exposure* to each source of risk. Cost uncertainty is inflationary when high-cost states remain adverse for the entrepreneur, while demand-level and demand-elasticity uncertainty are disinflationary under the corresponding net-exposure conditions. These opposing precautionary responses provide the basis for the markup-cyclicity results that follow.

3.3 Implications for markup cyclicity

The preceding results imply that different sources of uncertainty generate different precautionary pricing incentives. We now study how these pricing responses translate into markup cyclicity. As in the example of Section 2, it is useful to distinguish between the markup targeted by the firm when setting its price and the markup realized after shocks are observed. With non-constant marginal cost, this distinction is richer because realized shocks can affect marginal cost both directly and indirectly through their effect on the scale of production.

Ex-ante markups. Let $mc_{it}(p; \theta_{it}) := \mathcal{MC}_{it}(y_{it}(p; \theta_{it}); \theta_{it})$ denote realized marginal cost conditional on the posted price and the realization of shocks. For any posted price p , define the ex-ante markup as

$$\mu_{it}^{ea}(p) := \frac{p}{\mathbb{E}_{\theta_{it}}[mc_{it}(p; \theta_{it})]}.$$

The markup targeted by the firm under risk aversion α is therefore $\mu_{it}^{ea}(\alpha) := \mu_{it}^{ea}(p_{it}(\alpha))$, and the corresponding risk-neutral benchmark is $\mu_{it,RN}^{ea} := \mu_{it}^{ea}(p_{it,RN})$.

To characterize how a precautionary price adjustment translates into a markup adjustment, define

$$\Lambda_i := \left. \frac{\partial \mu_i^{\text{ea}}(p)}{\partial p} \right|_{(p, \theta) = (p_{i, \text{RN}}, 0)}.$$

The term Λ_i measures the local response of the ex-ante markup to the posted price at the risk-neutral benchmark. We evaluate Λ_i holding the underlying demand and cost structure fixed when studying time variation in uncertainty. Thus, Λ_i may differ across firms or sectors, reflecting differences in technologies and demand conditions.

Using the local pricing result, prices can be written as $p_{it}(\alpha) \approx p_{it, \text{RN}} + \alpha \mathcal{R}_{it}$. A first-order expansion of the ex-ante markup around the risk-neutral price then gives

$$\mu_{it}^{\text{ea}}(\alpha) \approx \mu_{it, \text{RN}}^{\text{ea}} + \alpha \Lambda_i \mathcal{R}_{it}.$$

Define the markup risk adjustment as $\mathcal{M}_{it} := \Lambda_i \mathcal{R}_{it}$. Then,

$$\mu_{it}^{\text{ea}}(\alpha) \approx \mu_{it, \text{RN}}^{\text{ea}} + \alpha \mathcal{M}_{it}.$$

With constant marginal cost, Λ_i reduces to the inverse of expected marginal cost. In the example of Section 2, where $\text{mc}_i = \bar{\kappa}_i + \eta_i$, $\Lambda_i = 1/\bar{\kappa}_i$, so the expression above reduces exactly to the first-order markup formula derived there. With non-constant marginal cost, Λ_i additionally captures the effect of the price-induced change in output on marginal cost.

Ex-post markups. The ex-post markup instead uses realized marginal cost. Once shocks are realized,

$$\mu_{it}^{\text{ep}} := \frac{p_{it}}{\text{mc}_{it}}.$$

The relationship between ex-ante and ex-post markups is exact:

$$\mu_{it}^{\text{ep}} = \mu_{it}^{\text{ea}} - \frac{\text{mc}_{it} - \mathbb{E}_{\theta_{it}}[\text{mc}_{it}]}{\text{mc}_{it}} \mu_{it}^{\text{ea}}.$$

The ex-post markup therefore reflects two distinct forces. The first is forward looking: firms adjust the markup they target in response to the distribution of shocks anticipated when the price is set. The second operates after the price has been chosen and reflects the realization of marginal cost relative to what the firm expected. With non-constant marginal cost, this realized marginal-cost component includes not only direct cost shocks but also movements in marginal cost induced by realized changes in output.

Aggregation. We next consider how firm-level markups map into sectoral and economy-wide markups. For any set of firms $\mathcal{I}$, define firm i 's sales share within that set as

$$\omega_{it} = \frac{p_{it}y_{it}}{\sum_{j \in \mathcal{I}} p_{jt}y_{jt}}.$$

The relevant aggregation set will always be clear from the summation: $\mathcal{I} = \mathcal{I}_{st}$ for sectoral aggregation and $\mathcal{I} = \mathcal{I}_t$ for economy-wide aggregation.

The harmonic sales-weighted markup for sector s is therefore

$$\mu_{st}^{\text{hsw}} = \sum_{i \in \mathcal{I}_{st}} \omega_{it} \mu_{it}^{-1}.$$

The corresponding ex-ante and ex-post sectoral markups satisfy

$$\mu_{st}^{\text{hsw,ep}} - \mu_{st}^{\text{hsw,ea}} = \sum_{i \in \mathcal{I}_{st}} \omega_{it} \frac{mc_{it} - \mathbb{E}_{\theta_{it}}[mc_{it}]}{p_{it}},$$

which makes clear that the relationship between ex-ante and ex-post aggregate markups depends on which realized marginal-cost shocks survive cross-sectional aggregation.

Proposition 3 (Aggregation of realized marginal-cost shocks). *Let the realized marginal-cost innovation of firm i be*

$$\widetilde{mc}_{it} := mc_{it} - \mathbb{E}_{\theta_{it}}[mc_{it}] = \lambda_i a_t + \varepsilon_{it},$$

where a_t is an aggregate shock, λ_i is a firm-specific exposure to that shock, and ε_{it} is an idiosyncratic innovation. Firms may differ arbitrarily in their marginal-cost functions, expected marginal costs, and exposures λ_i . Suppose that the idiosyncratic component satisfies a weighted cross-sectional law of large numbers

$$\sum_{i \in \mathcal{I}_{st}} \omega_{it} \frac{\varepsilon_{it}}{p_{it}} \rightarrow 0.$$

Then, in the large-sector limit,

$$\mu_{st}^{\text{hsw,ep}} - \mu_{st}^{\text{hsw,ea}} = a_t \sum_{i \in \mathcal{I}_{st}} \omega_{it} \frac{\lambda_i}{p_{it}}.$$

Hence, idiosyncratic realized marginal-cost innovations wash out under aggregation, whereas aggregate innovations generally remain.

Proof. The result follows directly from the definitions of the ex-ante and ex-post harmonic sales-weighted markups, the decomposition of realized marginal-cost innovations, and the weighted cross-sectional law of large numbers. $\square$

Proposition 3 allows firms to differ in both the level and shape of their marginal-cost functions. What matters for aggregation is not this technological heterogeneity itself, but whether realized marginal-cost innovations are diversifiable. Firm-specific innovations vanish in a sufficiently large cross-section, whereas shocks that move marginal costs systematically across firms survive.

The aggregate component a_t is a reduced-form marginal-cost innovation and need not correspond only to a primitive cost shock. With non-constant marginal cost, aggregate demand shocks may also generate a common marginal-cost component by changing firms' scale of production. Thus, both aggregate cost shocks and aggregate demand shocks can affect ex-post aggregate markups through realized marginal cost, while idiosyncratic components wash out under the conditions of the proposition.

Sectoral markup movements may also arise through reallocation across firms. Applying the standard midpoint identity to the inverse harmonic markup gives:

$$\Delta(\mu_{st}^{\text{hsw}})^{-1} = \underbrace{\sum_{i \in \mathcal{I}_{st}} \bar{\omega}_i \Delta \mu_{it}^{-1}}_{\text{within-firm markup changes}} + \underbrace{\sum_{i \in \mathcal{I}_{st}} \Delta \omega_{it} \bar{\mu}_i^{-1}}_{\text{within-sector reallocation}},$$

where bars denote the midpoint averages across periods.

The first term captures movements in firms' own markups holding sales shares fixed. The second captures shifts in activity across firms with different markups. Our mechanism provides a structural foundation for the first component by linking changes in firm-level markups to time-varying uncertainty.

The same logic extends to the economy-wide markup. The corresponding midpoint decomposition is

$$\Delta(\mu_t^{\text{hsw}})^{-1} = \underbrace{\sum_{s \in \mathcal{S}_t} \bar{\omega}_s \Delta(\mu_{st}^{\text{hsw}})^{-1}}_{\text{within-sector markup changes}} + \underbrace{\sum_{s \in \mathcal{S}_t} \Delta \omega_{st} (\bar{\mu}_{st}^{\text{hsw}})^{-1}}_{\text{between-sector reallocation}},$$

where $\mathcal{S}_t$ denotes the set of sectors.

Aggregate markup movements therefore combine changes in firms' own markups, reallocation across firms within sectors, and reallocation across sectors.

Our analysis adopts the standard notion of the aggregate markup used in the literature—the harmonic sales-weighted markup (see, e.g., [Baqae and Farhi, 2020](#); [Edmond et al., 2023](#); [Hasenzagl and Pérez, 2026](#); [Burstein et al., 2025](#)). The economic mechanism we emphasize, however, is distinct from that in [Burstein, Carvalho, and Grassi \(2025\)](#). In their framework, shocks change firms’ market shares and, through oligopolistic competition, their markups; these firm-level responses together with reallocation shape sector and aggregate markup fluctuations. In contrast, our framework shows how changes in uncertainty induce firms to adjust their ex-ante markups even holding market shares and market structure fixed. The two mechanisms are therefore complementary: market-share dynamics govern markup adjustment and aggregation in their framework, while precautionary pricing introduces an additional source of firm-level markup variation driven by changes in the distribution of risk.

Markup cyclicity. We now turn to the cyclicity of the economy-wide markup. Let Y_t denote aggregate economic activity. As in the example of Section 2, ex-ante markup cyclicity arises because the composition of uncertainty varies over the business cycle. Let $\sigma_{it}^2 = (\sigma_{\eta,it}^2, \sigma_{\xi,it}^2, \sigma_{\psi,it}^2)$ collect the variances of cost, demand-level, and demand-elasticity shocks faced by firm i in period t .

The aggregation decomposition above separates movements in firms’ own markups from reallocation within and across sectors. We first characterize how precautionary pricing contributes to aggregate ex-ante markup cyclicity, holding sales shares fixed.

Proposition 4 (Ex-ante aggregate markup cyclicity). *Suppose the assumptions of Theorem 2 and Proposition 1 hold for every firm and period, and let the distribution of shocks vary over time through σ_{it}^2 . Suppose further that $\mathcal{R}_{it}$ is continuously differentiable in σ_{it}^2 in a neighborhood of a reference uncertainty vector. Holding firm and sector sales shares fixed at their reference values, aggregate ex-ante markup cyclicity satisfies, to first order,*

$$\text{Cov}(\mu_t^{hsw,ea}, Y_t) \approx \text{Cov}(\mu_{t,RN}^{hsw,ea}, Y_t) + \alpha \sum_{i \in \mathcal{I}_t} \frac{(\bar{\mu}^{hsw,ea})^2 \bar{\omega}_i \Lambda_i}{(\bar{\mu}_i^{ea})^2} \sum_{k \in \{\eta, \xi, \psi\}} \underbrace{\frac{\partial \mathcal{R}_{it}}{\partial \sigma_{k,it}^2}}_{\text{exposure channel}} \times \underbrace{\text{Cov}(\sigma_{k,it}^2, Y_t)}_{\text{cyclical channel}},$$

where all derivatives are evaluated at the reference uncertainty vector, and bars denote values at the reference allocation.

Proof. See Appendix B.5. □

Proposition 4 shows that aggregate markup cyclicity reflects the interaction between pricing under risk and the cyclical behavior of uncertainty. Each source of risk

affects the precautionary component of markup dynamics through two distinct channels. The exposure channel, $\partial \mathcal{R}_{it} / \partial \sigma_{k,it}^2$, measures how sensitive optimal pricing is to uncertainty of type k , while the cyclical channel, $\text{Cov}(\sigma_{k,it}^2, Y_t)$, captures how that uncertainty co-moves with aggregate activity over the business cycle. Firm-level responses are mapped into the aggregate markup according to their contribution of the harmonic sales-weighted markup.

As a result, different risks need not have symmetric effects on aggregate markup dynamics. Even if multiple risks are equally volatile on average, their impact on markups depends on firms' exposure to each risk and how that risk co-moves with aggregate activity. Because cost uncertainty raises targeted markups while demand uncertainty lowers them, procyclical cost uncertainty and countercyclical demand uncertainty push markups toward procyclicality, whereas countercyclical cost uncertainty and procyclical demand uncertainty push them toward countercyclicality. Aggregate ex-ante markups may therefore be procyclical, countercyclical, or acyclical depending on the composition and cyclicity of risk.

We next turn to ex-post aggregate markup cyclicity.

Proposition 5 (Ex-post aggregate markup cyclicity). *Define the aggregate realized marginal-cost wedge as*

$$\mathcal{X}_t := \frac{\sum_{i \in \mathcal{I}_t} y_{it} (mc_{it} - \mathbb{E}_{\theta_{it}}[mc_{it}])}{\sum_{i \in \mathcal{I}_t} y_{it} mc_{it}}.$$

Then ex-post aggregate markup cyclicity admits the exact decomposition

$$\text{Cov}(\mu_t^{hsw,ep}, Y_t) = \underbrace{\text{Cov}(\mu_t^{hsw,ea}, Y_t)}_{\text{uncertainty channel}} - \underbrace{\text{Cov}(\mathcal{X}_t \mu_t^{hsw,ea}, Y_t)}_{\text{realized-cost channel}}.$$

Proof. See Appendix B.6. □

Proposition 5 shows that ex-post aggregate markup cyclicity reflects two forces. The uncertainty channel captures the cyclicity of the markup firms target when setting prices under cost and demand uncertainty, whereas the realized-cost channel captures the effect shocks have on the markup after prices have been set. In the constant-marginal-cost example of Section 2, $mc_t = \bar{\kappa} + \eta_t$, so $\mathcal{X}_t = \eta_t / (\bar{\kappa} + \eta_t)$, and the decomposition reduces exactly to the one obtained in the example.

3.4 Taking stock

The results of this section establish a general risk-management role for prices. Because changing the price changes profits differently across states, a risk-averse firm can use its price and, hence, its markup to redistribute resources across realizations of uncertainty. Cost uncertainty raises prices and markups, whereas demand uncertainty lowers them.

These pricing responses translate into markup cyclicalities when the distribution of uncertainty varies over time. Markup cyclicalities therefore depend not only on the amount of uncertainty firms face, but also on its composition and how each source of uncertainty comoves with aggregate activity. Procyclical cost uncertainty pushes markups toward procyclicalities, whereas procyclical demand uncertainty pushes them toward countercyclicalities. The cyclicalities of measured aggregate markups additionally reflect two forces that operate beyond firms' ex-ante pricing responses to uncertainty: the cyclical behavior of realized marginal cost shocks, which may reflect both cost and demand shocks, and the cyclical reallocation of sales across firms and sectors.

Ultimately, the extent to which uncertainty affects markup cyclicalities depends on the strength of the precautionary component of the markup, which hinges on three factors. First, greater risk aversion increases the strength of the pricing response to uncertainty. Second, higher pass-through strengthens this response by making it less costly, in expected-profit terms, to adjust prices away from the risk-neutral benchmark. Third, the response depends on how much of the firm's risk is ultimately borne by the entrepreneur. Non-profit resources and other risk-sharing arrangements can reduce this exposure. Partial insurance weakens the response of prices to a shock, while complete insurance eliminates it. More generally, even without explicit hedging, firm-level risk may have little effect on pricing when fluctuations in profits are small relative to the entrepreneur's wealth. Appendix C specializes these results to canonical monopolistic-pricing environments with linear and isoelastic demand under CARA and CRRA preferences, providing closed-form illustrations of the general mechanisms and showing how demand curvature, pass-through, and wealth-dependent risk aversion shape the quantitative response of markups to different sources of uncertainty.

4 Empirical Validation and Sources of Markup Cyclicalities

This section tests the theory's central predictions directly on markups. The theory implies that different sources of uncertainty have opposing effects on ex-ante markups: cost uncertainty raises them, whereas demand uncertainty lowers them. Under the conditions of Proposition 1, these sign predictions hold under strictly concave preferences

and for a broad class of demand systems, although the magnitude of each response varies across economic environments.

We proceed in four steps. First, we measure ex-post industry markups and construct industry-specific, time-varying measures of cost and demand uncertainty from each industry's position in the production network. Second, we describe our empirical strategy and sources of identifying variation. Third, we estimate the response of markups to each source of uncertainty, instrumenting our uncertainty measures with leave-one-out shift-share instruments based on second moments. We do so using both our baseline industry sample from the NBER-CES data and a firm-level sample from U.S. Compustat. In the firm-level analysis, we ask whether a given firm changes its markup when uncertainty in its industry changes, thereby distinguishing changes in firms' pricing behavior from changes in industry composition. Finally, using the estimated coefficients in our baseline sample, we decompose observed markup cyclicity into the contributions of uncertainty and realized cost and demand conditions.

4.1 Data and Measurement

Our empirical analysis combines industry-level data from the NBER-CES Manufacturing Industry Database with input-output linkages from the BEA Benchmark Detail Use Tables. We measure markups using the cost-share approach of [De Loecker and Warzynski \(2012\)](#). We then construct industry-specific measures of cost and demand uncertainty from forecast surprises in supplier-cost and customer-demand conditions, weighted by each industry's position in the production network.

Industry panel. We use the NBER-CES Manufacturing Industry Database, which provides annual data on shipments, payroll, materials expenditure, capital stocks, and price deflators for 6-digit NAICS manufacturing industries from 1958 to 2018.⁴ The estimation panel is restricted to years covered by the IO-table linkages and by the burn-in required for the uncertainty measure described below. The resulting sample contains 358 manufacturing industries over 1972–2018 and 16,751 industry-year observations.

Measuring markups. The markup is measured with the cost-share approach of [De Loecker and Warzynski \(2012\)](#): for a variable input X , $\mu_{jt} = v_{X,j}/\alpha_{X,jt}$, where $v_{X,j}$ is the output elasticity of X and $\alpha_{X,jt} = \text{cost}_{X,jt}/\text{sales}_{jt}$ its revenue share. We implement

⁴We use the 2021 vintage, which provides consistent industry definitions based on the 2012 NAICS classification. See [Bartelsman and Gray \(1996\)](#) for documentation.

this approach for the flexible *bundle* of labor and materials:

$$\mu_{jt} = \frac{\nu_{L,j} + \nu_{M,j}}{\alpha_{L,jt} + \alpha_{M,jt}}, \quad \alpha_{X,jt} = \frac{\text{cost}_{X,jt}}{\text{sales}_{jt}}. \quad (1)$$

Production is Cobb–Douglas with constant returns over labor, materials, and capital, so each input’s output elasticity equals its share of total cost. We set the elasticities to common values across industries, $\theta_L = 0.3$ for labor and $\theta_M = 0.6$ for materials, implying a labor-plus-materials elasticity of 0.90.⁵ This implementation does not require production-function estimation.⁶ Because the markup measure contains realized variable cost, the measured markup is the ex-post markup, not the ex-ante markup that is directly determined by firms’ pricing decisions under uncertainty. Section 4.2 explains how our empirical design isolates the ex-ante pricing response.

Production-network exposure. The theory predicts that markups respond to cost and demand uncertainty. To measure the cost and demand conditions faced by each industry, we first construct input-output shares using the BEA Benchmark Detail Use Tables. The data contain ten benchmark vintages spanning 1972–2017, released every five years in years ending in 2 and 7.

We classify industries using the NAICS-based input-output classification of the 2007, 2012, and 2017 tables, and map the earlier tables to that classification. Tables through 1992 classify industries by BEA IO codes defined over SIC industries, whose classification changes across vintages. We use BEA sectoring plans and vintage bridges to map the earlier tables to SIC-87. The plans and bridges are code lists without weights, so we use them only to determine which SIC-87 industries each IO code contains; we then construct weights at the SIC-87 level, using the concordance of [Fort and Klimek \(2018\)](#), built from Census establishment records observed under both classifications. Each SIC-87 industry in an IO code’s set is weighted by its shipments, and an SIC industry that maps to several NAICS industries is split across them in proportion to the shipments of the establishments in each SIC–NAICS pair. We then use BEA’s NAICS-to-IO concordance to map the NAICS industries to the input-output classification of the 2007, 2012, and 2017 tables. Between benchmark table years, we linearly interpolate the detailed use shares to obtain annual input–output weights; after 2017, the weights

⁵Pricing capital at a Hall–Jorgenson rental rate, built from the Moody’s Baa yield, expected investment-price inflation, and depreciation from each industry’s equipment/structures mix, and computing each industry’s time-averaged shares of total cost gives a labor-plus-materials share of 0.91 at the sales-weighted mean, 0.92 at the median, and 0.86–0.96 across the 10th–90th percentiles of industries.

⁶This construction avoids the identification concern emphasized by [Bond et al. \(2021\)](#): when only revenue data are observed, an estimated “output elasticity” is a revenue elasticity and cannot by itself identify the markup level. The cost-share construction uses only accounting shares and a constant-returns assumption. Labor-only and materials-only versions of the markup provide independent measures of the same object and co-move closely; the bundle is our baseline throughout.

are held at their 2017 values, affecting only 2018. For each manufacturing industry j , we construct supplier shares $s_{jk,t}$, defined as the share of industry j 's intermediate purchases supplied by industry k , and customer shares $w_{jk,t}$, defined as the share of industry j 's sales purchased by customer k .

Cost and demand conditions. The *cost condition* facing industry j is the supplier-share-weighted growth rate of its suppliers' output prices:

$$C_{jt} = \sum_{k: n_3(k) \neq n_3(j)} s_{jk,t} \Delta \log P_{kt},$$

where $s_{jk,t}$ is the supplier share constructed from the input-output tables and $n_3(k)$ denotes industry k 's three-digit NAICS sector. P_{kt} denotes supplier k 's output price. For manufacturing suppliers, we use the industry's NBER-CES shipments price index; for non-manufacturing suppliers, we use commodity producer-price indexes where a close match exists—fuels and power for energy suppliers, metals for mining, and farm products for agriculture—and the GDP deflator for services and the remaining sectors.

The supplier shares come from the same Use tables, restricted to intermediate purchases: final demand buys output but supplies no inputs, so only industries appear as suppliers. Cross-industry variation in the resulting cost condition therefore comes from differences in network exposure as well as, for manufacturing suppliers, differences in supplier-specific price movements.

The *demand condition* facing industry j is the customer-share-weighted growth rate of its buyers' real activity:

$$D_{jt} = \sum_{k: n_3(k) \neq n_3(j)} w_{jk,t} \Delta \log A_{kt},$$

where $w_{jk,t}$ is the customer share constructed from the input-output tables and A_{kt} is buyer k 's real activity. For manufacturing buyers, we use real shipments from the NBER-CES; for final-demand categories and non-manufacturing buyers, we use the corresponding national-accounts aggregates.

Final demand enters the demand condition like any other buyer. The Use tables record each industry's sales to final users, which we sum into four categories: personal consumption, private investment (fixed investment plus inventory changes), exports, and government. Imports enter the tables negatively and are excluded. Each category's activity series is the corresponding chain-type quantity index of NIPA Table 1.1.3; non-manufacturing industry buyers are assigned the NIPA index matching their sector,

for example personal consumption of health-care services for health-care buyers and residential investment for residential construction.

The conditions are measured at the detail IO level, one step coarser than the six-digit NAICS level of the markups: 213 manufacturing IO industries in the 2007 classification. For manufacturing counterparties, prices and activity are built by aggregating the six-digit NBER-CES data to the IO industry, using the implicit price deflator and the sum of real shipments of its member industries; non-manufacturing and final-demand counterparties carry the sectoral and national series described above. Each six-digit NAICS industry is assigned the conditions of the IO industry containing it.

Two features of the construction are important for identification. First, both conditions exclude counterparties in the industry's own three-digit NAICS sector: suppliers on the cost side and industrial buyers on the demand side. Final-demand categories, which belong to no NAICS sector, are never excluded, and the weights are renormalized to sum to one after the exclusion. This restriction reduces the mechanical relationship between an industry's own pricing and activity and its measured cost and demand conditions, and it is similar in spirit to the shift-share designs of [Goldsmith-Pinkham, Sorkin, and Swift \(2020\)](#). Second, both conditions interact shocks to counterparties with predetermined differences in network exposure. For example, an increase in a supplier's price affects every industry that purchases from that supplier, but the magnitude of the exposure varies with the supplier's share in each industry's input bundle. This interaction between common shocks and heterogeneous network positions generates the variation used below.

Surprises and uncertainty. The theoretical object is the conditional variance of cost and demand shocks at the time prices are set. We construct empirical counterparts in two stages, using only information available before the period in which the markup is measured. First, for each industry we compute *surprises* from an expanding-window AR(1) forecast of the cost condition C_{jt} and the demand condition D_{jt} . This follows the forecast-based approach of the uncertainty-measurement literature ([Jurado et al., 2015](#); [Ludvigson et al., 2021](#)): uncertainty is the volatility of unforecastable innovations, so predictable variation in the conditions is removed before volatility is measured. The surprise is the component of the current condition that the industry's own history could not predict:

$$u_{S,jt} = S_{jt} - \widehat{b}_{0,jt}^S - \widehat{b}_{1,jt}^S S_{j,t-1}, \quad S \in \{C, D\}, \quad (2)$$

where $(\widehat{b}_{0,j,t}^S, \widehat{b}_{1,j,t}^S)$ are least-squares estimates of the AR(1) on industry j 's own history through year $t - 1$, re-estimated each year as the sample expands; the first surprise is computed once at least eight annual observations are available.

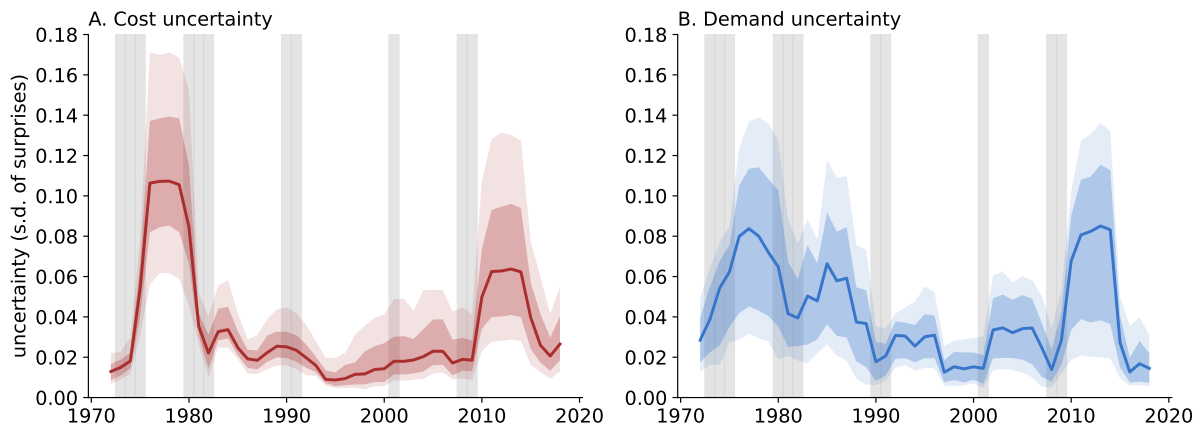
Second, we measure *uncertainty* as the trailing standard deviation of the five most recent surprises available before year t :

$$\widehat{\sigma}_{S,j,t-1} = \text{SD}(u_{S,j,t-5}, \dots, u_{S,j,t-1}), \quad S \in \{C, D\}. \quad (3)$$

The date- t uncertainty measure is therefore predetermined with respect to the date- t markup: it uses only surprises realized before the year whose markup it explains. The window stops at $t - 1$ because the theory's object is the ex-ante markup while the measured markup is the ex-post markup and therefore also reflects the date- t realization of marginal cost. Stopping at $t - 1$ prevents the uncertainty measure from mechanically incorporating the same realization that enters the measured markup; Section 4.2 develops this point.

The measures vary both across industries, because industries occupy different positions in the production network, and over time, because the volatility of supplier-price and customer-activity surprises changes. Figure 4 plots the cross-industry distribution of each measure over time. Both cost and demand uncertainty rise around the oil shocks of the 1970s and early 1980s and the 2008–09 financial crisis. Because the measures use trailing windows, their peaks occur shortly after the largest underlying surprises. Both measures also exhibit substantial cross-industry dispersion. Demand uncertainty is persistently dispersed across industries, whereas cost uncertainty becomes especially dispersed during the oil-shock era and after 2008.

Figure 4: Cost and demand uncertainty over time.



Notes: The distribution of the trailing uncertainty measures of equation (3) across input-output industries, per year, in units of the standard deviation of surprises: the median (line), the interquartile range (dark band), and the 10th–90th percentile range (light band), 1972–2018. Gray bands are NBER recessions.

4.2 Empirical Strategy and Identification

Our main specification relates industry-level log markups to predetermined uncertainty and predetermined first-moment controls:

$$\log \mu_{jt} = \gamma_{\text{cost}} \widehat{\sigma}_{C,j,t-1} + \gamma_{\text{demand}} \widehat{\sigma}_{D,j,t-1} + \beta_C \overline{C}_{j,t-1} + \beta_D \overline{D}_{j,t-1} + \delta_j + \varepsilon_{jt}. \quad (4)$$

The timing of each variable matters, so we state it explicitly. The dependent variable is the ex-post markup in year t : it is measured from realized revenue and cost and therefore incorporates the year's realized shocks. The uncertainty measures $\widehat{\sigma}_{C,j,t-1}$ and $\widehat{\sigma}_{D,j,t-1}$ are constructed from surprises through $t-1$, as in equation (3). The controls $\overline{C}_{j,t-1}$ and $\overline{D}_{j,t-1}$ are the trailing five-year means of the cost and demand conditions through $t-1$, using the same window as the uncertainty measures; they separate the response to uncertainty from the response to the first moments of the underlying conditions. δ_j denotes industry fixed effects. The dependent variable is therefore the only date- t object in the equation: every regressor is predetermined. All four regressors are divided by their standard deviation over the full panel, so each coefficient reports the response to a one-standard-deviation change.

This specification is the direct empirical counterpart of Proposition 1. The theory predicts $\gamma_{\text{cost}} > 0$ and $\gamma_{\text{demand}} < 0$. Greater cost uncertainty raises the ex-ante markup, whereas greater demand uncertainty lowers it. Under risk neutrality, markups may respond to realized cost and demand conditions but not to their conditional volatility, implying $\gamma_{\text{cost}} = \gamma_{\text{demand}} = 0$. The opposing signs therefore provide a joint test of the precautionary-pricing mechanism. In particular, explanations based on common shifts in market power or demand elasticity do not naturally predict opposite markup responses to cost and demand uncertainty.

Identification. Trailing volatility measures are noisy measures of conditional uncertainty: each is a standard deviation of only five estimated surprises, and measurement error would attenuate the least-squares estimate toward zero. We therefore instrument each uncertainty measure with a leave-one-out shift-share second-moment instrument:

$$\begin{aligned} Z_{j,t-1}^C &= \sum_{k: n_3(k) \neq n_3(j)} \bar{s}_{jk} \text{SD}(\Delta \log P_{k,t-5}, \dots, \Delta \log P_{k,t-1}), \\ Z_{j,t-1}^D &= \sum_{k: n_3(k) \neq n_3(j)} \bar{w}_{jk} \text{SD}(\Delta \log A_{k,t-5}, \dots, \Delta \log A_{k,t-1}), \end{aligned}$$

where $\bar{s}_{jk}$ and $\bar{w}_{jk}$ are the sample averages of the annual shares $s_{jk,t}$ and $w_{jk,t}$, renormalized to sum to one, and the counterparties, supplier prices P , and buyer activities A are those of the conditions in Section 4.1. Each instrument is predetermined, its window

ending at $t - 1$, and because its weights are fixed and counterparties in the industry's own three-digit sector are excluded, it moves with volatility among the industry's suppliers and customers rather than directly with the industry's own shocks.

The level controls $\bar{C}_{j,t-1}$ and $\bar{D}_{j,t-1}$ have the analogous first-moment instruments: the same fixed weights applied to the mean of the same windows,

$$\begin{aligned} Z_{j,t-1}^{\bar{C}} &= \sum_{k: n_3(k) \neq n_3(j)} \bar{s}_{jk} \text{Mean}(\Delta \log P_{k,t-5}, \dots, \Delta \log P_{k,t-1}), \\ Z_{j,t-1}^{\bar{D}} &= \sum_{k: n_3(k) \neq n_3(j)} \bar{w}_{jk} \text{Mean}(\Delta \log A_{k,t-5}, \dots, \Delta \log A_{k,t-1}). \end{aligned}$$

Each instrument combines fixed shares of an industry's suppliers or customers with time variation in the volatility of those suppliers or customers, yielding the shift-share structure studied by [Goldsmith-Pinkham et al. \(2020\)](#), [Adao et al. \(2019\)](#), and [Borusyak et al. \(2022\)](#). Inference is two-way clustered by industry and year. Our baseline specification omits year fixed effects. The object of the paper is markup cyclical, and the theory predicts that pricing responds to uncertainty including its aggregate component. Year fixed effects would absorb this common time variation and instead identify the response to differences in uncertainty across industries within a given year. That specification therefore answers a distinct question from the baseline rather than providing a strictly more demanding version of the same test. Table 5 reports the year-effects specification together with other robustness exercises.

4.3 Empirical Evidence

Table 1 presents the main results. Column (2) reports the instrumental-variable estimates of equation (4). Markups respond to cost and demand uncertainty with the signs predicted by the theory: $\gamma_{\text{cost}} = +1.3$ and $\gamma_{\text{demand}} = -1.2$. The coefficients are measured in log points per one-standard-deviation increase in uncertainty. The first-stage F -statistics are 53 for cost uncertainty and 440 for demand uncertainty. Ordinary least squares, reported in column (1), yields the same opposing signs but coefficients approximately half as large, a pattern consistent with attenuation from measurement error in the generated uncertainty measures. Column (3) instruments the controls $\bar{C}_{j,t-1}$ and $\bar{D}_{j,t-1}$ as well, using the first-moment instruments of Section 4.2; the estimates are similar. The estimated responses retain the predicted opposing signs across additional specifications reported in Appendix D, including downside-risk measures of uncertainty and the subsample of high-materials-share industries in which the markup is most cleanly measured.

Table 1: Industry-level markup responses to cost and demand uncertainty.

Dependent variable: log markup	(1) OLS	(2) IV	(3) IV
Cost uncertainty, $\widehat{\sigma}_{C,jt}$	+0.5 (+1.4)	+1.3* (+1.9)	+1.3** (+2.0)
Demand uncertainty, $\widehat{\sigma}_{D,jt}$	-0.8* (-1.9)	-1.2** (-2.1)	-1.2** (-2.0)
First-moment controls (cost, demand)	✓	✓	✓
First moments instrumented	—	—	✓
Industry fixed effects	✓	✓	✓
First-stage F (cost / demand)	—	53/440	50/392
Observations	16,751	16,751	16,751

Notes. The table reports estimates based on NBER–CES manufacturing data from 1972–2018. The dependent variable is the log markup. Cost and demand uncertainty are standardized over the full sample, so coefficients report the response to a one-standard-deviation increase in uncertainty, in log points (coefficient $\times 100$). All specifications control for the trailing means of the cost and demand conditions. Column (2) instruments each uncertainty measure with its own-sector-excluded shift-share second-moment instrument. Column (3) additionally instruments the first-moment controls. Parentheses report t -statistics based on two-way clustering by industry and year. * $p < .10$, ** $p < .05$, *** $p < .01$.

Firm-level evidence. An industry’s markup can respond to uncertainty through two distinct channels: incumbent firms may change their markups, as in the theory, or the composition of operating firms may shift toward producers with systematically higher or lower markups. The harmonic-sales-weighted decompositions of Section 3 distinguish these channels conceptually; firm-level data allow us to distinguish them empirically. We use Compustat manufacturing firms from 1972 through 2018 and merge them with the industry uncertainty measures of Section 4.1 by NAICS industry and year. The resulting panel contains 102,689 firm-year observations for 7,188 firms.

We construct firm-level markups using the cost-share approach, as in equation (1), using cost of goods sold as the flexible-input bundle. Cost of goods sold includes direct materials, direct labor, and factory overhead and is therefore the closest Compustat counterpart to the labor-and-materials bundle used in the industry analysis. Because the output elasticity ν is assumed constant within an industry, it is absorbed by the firm fixed effect for firms that remain in the same industry. The within-firm estimates therefore do not require us to separately identify its level.

We estimate the same specification as equation (4), replacing industry fixed effects with firm fixed effects. We use the same uncertainty measures, shift-share instruments, and first-moment controls, and winsorize the markup measure at the 1st and 99th percentiles. The harmonic sales-weighted aggregate markup is close to 1.30 in both datasets, and in both it rises over the sample, from about 1.20 in the early 1970s to about 1.35 by 2018. Table 2 reports the results. Within firms, the estimated responses have the signs predicted by the theory. The demand-uncertainty coefficient is -1.0 log points,

with a t -statistic of -2.0 , close to the industry-level response of -1.2 in Table 1. The difference between the cost- and demand-uncertainty coefficients is 1.7 log points, with a t -statistic of 1.9. The estimates reject equality of the two responses at the ten-percent level and are consistent with the theory's prediction that cost and demand uncertainty affect markups in opposite directions. The within-firm estimates retain the predicted signs under two-way clustering, without winsorization, and with the broader cost bundle that adds SG&A to cost of goods sold, as shown in Table 6 of Appendix D.

Table 2: Firm-level markup responses to industry uncertainty.

Dependent variable: log markup	Cost uncertainty	Demand uncertainty	Obs.
<i>Within firms</i> (firm fixed effects)	+0.7 (+1.3)	-1.0** (-2.0)	102,689
<i>Across firms</i> (industry fixed effects)	+0.1 (+0.3)	-0.3 (-0.8)	102,689
<i>Within firms, by firm size</i>			
Q1 (smallest)	+1.4 (+0.6)	-0.7 (-0.4)	15,817
Q2	+2.5 (+1.6)	-1.6** (-2.0)	23,128
Q3	+1.4* (+1.7)	-1.8*** (-3.0)	26,827
Q4 (largest)	-0.1 (-0.4)	-0.4 (-1.0)	36,917
Top decile	-0.8** (-2.4)	-0.7* (-1.7)	16,744

Notes. The table reports estimates based on manufacturing firms from U.S. Compustat for 1972–2018. All specifications are estimated by two-stage least squares, instrumenting cost and demand uncertainty with the second-moment shift-share instruments of Section 4.2 and controlling for the first moments of the cost and demand conditions. The dependent variable is the log markup, winsorized at the 1st and 99th percentiles. The lower panel splits the within-firm specification by quartiles of firm size, measured by mean firm sales over the sample, and reports the top sales decile separately. Each quartile contains 1,797 firms and the top decile 719 firms. Coefficients report the response to a one-standard-deviation increase in uncertainty, in log points (coefficient $\times 100$). Parentheses report t -statistics clustered by year. * $p < .10$, ** $p < .05$, *** $p < .01$.

The comparison between firm- and industry-fixed-effect specifications is informative. When the same Compustat panel is estimated with industry rather than firm fixed effects, the coefficients are close to zero, suggesting that compositional changes among publicly listed firms attenuate the within-firm relationship in industry aggregates. The within-firm results therefore show that changes in continuing firms' markups contribute to the industry-level relationship in Table 1 and rule out an explanation based entirely on compositional change.

The lower panel of Table 2 shows how the response varies with firm size. The point estimates are at or above the industry-level slopes for small and mid-sized firms,

attenuate toward zero in the largest quartile, and turn slightly negative for cost uncertainty in the top sales decile. This gradient is consistent with greater diversification, hedging, and access to financial markets reducing the effective risk aversion relevant for pricing among larger firms, although greater scale also increases the amount of profit exposed to uncertainty. The NBER-CES data cover the broader manufacturing universe, including privately held firms whose owners may be less diversified than those of publicly listed firms. The concentration of the Compustat response among smaller firms therefore suggests that the within-firm mechanism may remain quantitatively important outside the sample of public firms.

4.4 An Empirical Decomposition of Markup Cyclicity

We now ask how much of observed aggregate markup cyclicity can be attributed to the channels estimated above. The regression separates two types of information available when firms set prices. The uncertainty measures capture variation in the second moments of cost and demand conditions, while the first-moment controls capture their predictable component. Both are constructed using information available through year $t - 1$. By contrast, cost and demand shocks realized during year t are not included in these regressors. Their effects on measured ex-post markups therefore appear in the residual, together with measurement error and other components not captured by the regression.

Using the estimated coefficients from our baseline specification, we construct the contribution of uncertainty and predictable cost and demand conditions to aggregate markup fluctuations over time. The resulting decomposition is:

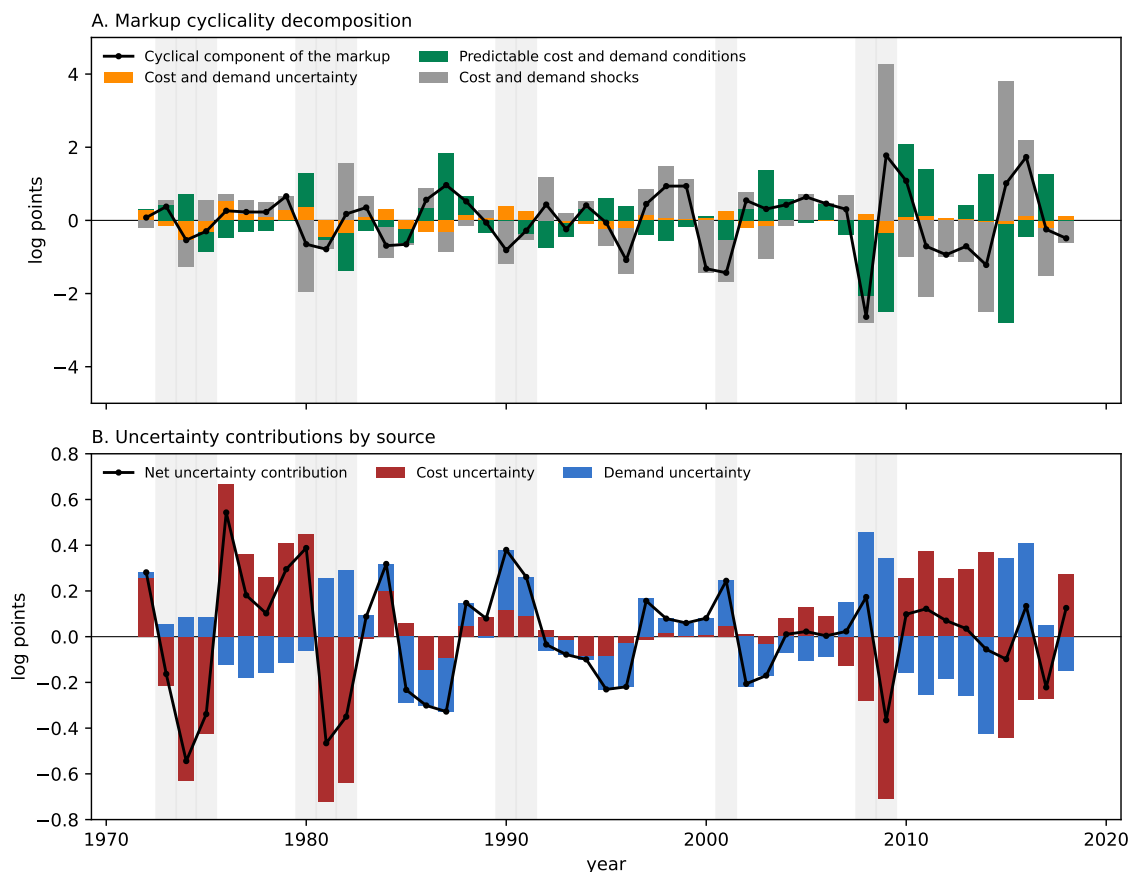
$$\log \mu_t^{\text{cyc}} = \underbrace{\widehat{\gamma}_{\text{cost}} \widehat{\sigma}_{C,t-1}^{\text{cyc}} + \widehat{\gamma}_{\text{demand}} \widehat{\sigma}_{D,t-1}^{\text{cyc}}}_{\text{cost and demand uncertainty}} + \underbrace{\widehat{\beta}_C \overline{C}_{t-1}^{\text{cyc}} + \widehat{\beta}_D \overline{D}_{t-1}^{\text{cyc}}}_{\text{predictable cost and demand conditions}} + \underbrace{r_t}_{\text{cost and demand shocks}} \quad (5)$$

The coefficients are the instrumental-variable estimates from column (2) of Table 1. We aggregate the industry-level contributions using the weights implied by the harmonic sales-weighted markup and extract cyclical components using the Hodrick–Prescott filter with $\lambda = 6.25$, following [Ravn and Uhlig \(2002\)](#).

Figure 5 shows that uncertainty accounts for a sizable component of aggregate markup cyclicity, particularly in the earlier part of the sample. Its contribution is especially important before the Great Moderation (mid 1980s), when uncertainty-induced movements in markups are comparable in magnitude to the observed markup cycle in several episodes. The contribution of uncertainty becomes smaller in the later part of the sample, when a larger share of aggregate markup fluctuations is accounted for

by the realized cost and demand shocks. The composition of uncertainty also varies sharply across episodes. During the oil-shock period, the uncertainty contribution is dominated by cost risk, whereas demand uncertainty becomes relatively more important in later downturns and accounts for much of the uncertainty contribution around the 2001 and 2008–10 recessions.

Figure 5: Decomposition of the cyclical component of the aggregate markup.



Notes. *Panel A* decomposes the cyclical component of the aggregate NBER-CES markup (line) into the contributions of uncertainty, predictable cost and demand conditions, and the residual (bars). *Panel B* decomposes the uncertainty contribution (line) into cost and demand components (bars). Gray bands indicate NBER recessions.

5 Quantitative Model

We now embed the precautionary-pricing mechanism in a dynamic general-equilibrium model of firm entry and exit, building on [Hopenhayn \(1992\)](#) and [Lucas \(1978\)](#). Heterogeneous firms compete under monopolistic competition. They post prices before current cost and demand shocks are realized and commit to satisfying demand at those prices, so pricing remains an instrument for managing within-period profit risk. They also decide whether to enter and whether to continue operating, so the same

uninsurable risk that moves prices shapes firm selection and the equilibrium mass of varieties. Free entry determines the toughness of competition.

We discipline the model using four moments: the two markup responses to cost and demand uncertainty, the average markup, and the firm exit rate. We use the model to quantify the welfare value of precautionary pricing when entry, selection, competition, and product variety all adjust.

The model connects the theory, measurement, and counterfactuals through a common set of economic objects. Productivity and additive demand shocks correspond to the cost and demand innovations measured in Section 4, while their firm, industry, and aggregate components distinguish risks that can be pooled from risks borne by the economy as a whole. Labor and materials enter through the same variable-cost aggregate used to construct empirical markups, and decreasing returns capture quasi-fixed inputs. CARA utility makes absolute risk aversion independent of wealth and allows the estimated risk-aversion parameter to summarize effective exposure to unshared profit risk, combining primitive risk aversion with the extent of financial risk sharing. Because agents also consume the differentiated goods produced in the model, aggregate changes in differentiated-goods surplus enter their consumption index. This channel matters for aggregate risk and welfare. The theoretical mechanism itself holds for general concave utility. The model abstracts from nominal rigidities, customer capital, and other motives for price adjustment. This structure makes the markup responses discipline the strength of the mechanism and gives the welfare counterfactuals a transparent interpretation.

5.1 Environment and Equilibrium

Agents and sectors. Time is discrete, $t = 0, 1, 2, \dots$. The economy is populated by a unit mass of ex-ante identical agents and contains two production sectors. A competitive outside sector produces a homogeneous numeraire good. In the differentiated sector, each *entrepreneur* operates a firm producing a distinct variety. Agents who do not operate firms are *workers*. All agents supply $\bar{\ell}$ units of labor inelastically and earn labor income $w_t \bar{\ell}$. Entrepreneurs additionally receive the realized profits of their firms. Occupational status is endogenous: workers may enter entrepreneurship by paying a sunk cost, while incumbent entrepreneurs may exit and become workers. There is no asset market, so every agent consumes its resources each period and, conditional on operating, pricing is the only within-period instrument for managing exposure to firm-profit risk; Appendix E.1 shows that savings reduce effective risk aversion and attenuate the quantitative importance of precautionary pricing, while leaving its qualitative predictions unchanged.

Firms are grouped into industries indexed by j . Industries are the units at which uncertainty and markups are measured in Section 4, but they are not separate product markets. In the model, all firms compete through a common differentiated-goods demand system. Industry membership instead determines firms' common exposure to cost and demand shocks.

Cost and demand processes. Each firm is described by a productivity level z_{it} and a demand shifter ξ_{it} , which enter the technology and the demand curve below.⁷ Each contains an aggregate component and an industry-firm component:

$$\begin{aligned}\log z_{it} &= Z_t + \tilde{z}_{it}, \\ \xi_{it} &= D_t + \tilde{\xi}_{it}.\end{aligned}\tag{6}$$

The aggregate cost and demand states evolve according to

$$\begin{aligned}Z_t &= \rho_Z Z_{t-1} + \sigma_{c,t} \Omega_{c,t}, \\ D_t &= \rho_D D_{t-1} + \sigma_{d,t} \Omega_{d,t},\end{aligned}\tag{7}$$

where $\Omega_{c,t}$ and $\Omega_{d,t}$ are standardized aggregate innovations. The conditional standard deviations $\sigma_{c,t}$ and $\sigma_{d,t}$ vary over time and constitute aggregate cost and demand uncertainty.

The industry-firm components evolve as

$$\begin{aligned}\tilde{z}_{it} &= \rho_z \tilde{z}_{it-1} + \sigma_{z,jt} \omega_{z,it} + \sigma_{z,\text{idio}} \varepsilon_{z,it}^0, \\ \tilde{\xi}_{it} &= \rho_\xi \tilde{\xi}_{it-1} + \sigma_{\xi,jt} \omega_{\xi,it},\end{aligned}\tag{8}$$

The terms $\sigma_{z,jt}$ and $\sigma_{\xi,jt}$ are time-varying industry cost and demand uncertainty. The component $\sigma_{z,\text{idio}} \varepsilon_{z,it}^0$ generates persistent firm heterogeneity and provides a time-invariant floor of productivity dispersion. It helps determine the firm-size distribution and the exit rate but does not generate time variation in uncertainty.

Firms in the same industry share part of their innovations. For $k \in \{z, \xi\}$, write

$$\omega_{k,it} = \kappa_k \zeta_{k,jt} + \sqrt{1 - \kappa_k^2} \varepsilon_{k,it},$$

⁷These are the shocks written generically as θ in Section 3, so that $\theta_{it} = (z_{it}, \xi_{it})$ here. The stylized example of Section 2 instead writes the cost shock as an additive shifter of a constant marginal cost, $\kappa(\eta) = \bar{\kappa} + \eta$; its counterpart here is realized marginal cost, $\bar{\kappa} + \eta_{it} = \mathcal{C}_y(y_{it}; z_{it})$, given below. Decreasing returns introduce two differences. Productivity shifts marginal cost multiplicatively rather than additively: at a given output, a one percent rise in z_{it} lowers marginal cost by $1/\nu$ percent. And marginal cost varies with realized output, so the demand shifter moves it as well. Both differences vanish at $\nu = 1$, where $\bar{\kappa} + \eta$ corresponds exactly to w_t/z_{it} .

where $\zeta_{k,jt}$ is common to firms in industry j , while $\varepsilon_{k,it}$ is specific to firm i . Both innovations have unit variance and are mutually independent. The parameter $\kappa_k^2 \in [0, 1)$ is the correlation between the shocks of two firms in the same industry and determines the fraction of industry uncertainty that survives aggregation to the industry markup.

Conditional cost uncertainty consequently decomposes as

$$\text{Var}_{t-1}(\log z_{it}) = \underbrace{\sigma_{c,t}^2}_{\text{aggregate}} + \underbrace{\sigma_{z,jt}^2}_{\text{industry}} + \underbrace{\sigma_{z,\text{idio}}^2}_{\text{firm}}, \quad (9)$$

with an analogous decomposition on the demand side, except that there is no separate time-invariant idiosyncratic demand component.

The aggregate and industry uncertainty states follow autoregressive processes in logs. For cost uncertainty,

$$\begin{aligned} \log \sigma_{c,t}^2 &= (1 - \rho_c) \log \bar{\sigma}_c^2 + \rho_c \log \sigma_{c,t-1}^2 + \sigma_{vc} \eta_{c,t}, & (\text{aggregate}) \\ \log \sigma_{z,jt}^2 &= (1 - \rho_u) \log \bar{\sigma}_z^2 + \rho_u \log \sigma_{z,j,t-1}^2 + \sigma_{vu} \nu_{z,jt}. & (\text{industry}) \end{aligned} \quad (10)$$

Demand uncertainty follows analogous processes.

Industry uncertainty is independent across the continuum of industries, so it washes out in the aggregate. Its variation across industries and over time allows the model to reproduce the empirical markup–uncertainty regressions. Aggregate uncertainty, by contrast, is common to all firms, survives aggregation, and drives the business cycle.

Preferences and demand. Agents have time-additive preferences

$$\mathcal{U}_{it} = \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s U(c_{0,it+s} + \mathcal{V}(\{c_{k,t+s}\})), \quad (11)$$

where $\beta \in (0, 1)$ is the discount factor and utility over the consumption index is CARA:

$$U(x) \equiv -\alpha \exp(-\alpha x).$$

The parameter $\alpha > 0$ is the coefficient of absolute risk aversion. The differentiated-goods sub-utility follows [Melitz and Ottaviano \(2008\)](#):

$$\mathcal{V}(\{c_k\}) = \int \left[\frac{(a + \xi_k) c_k}{b} - \frac{c_k^2}{2b} \right] dk - \frac{\eta_Q}{2b} \left(\int c_k dk \right)^2,$$

where $a, b, \eta_Q > 0$. The shifter ξ_k captures demand for variety k , while η_Q governs the strength of competition across varieties. The shifter enters additively, which places

the model strictly inside the demand-risk case of Proposition 2; a multiplicative shifter would leave no demand-uncertainty response at all.

It is useful to express the consumption index in terms of the surplus generated by differentiated consumption. Define

$$\mathcal{S}_t := \mathcal{V}(\{c_{kt}\}) - \int p_{kt} c_{kt} dk$$

as per-capita differentiated-goods surplus. Because all agents face the same prices and have quasi-linear preferences over differentiated consumption, they choose the same differentiated-goods bundle. Using the household budget constraint, the consumption index of a worker can therefore be written as $w_t \bar{\ell} + \mathcal{S}_t$ while that of entrepreneur i is $w_t \bar{\ell} + \pi_{it} + \mathcal{S}_t$. Thus entrepreneurs bear firm-profit risk but, like workers, also receive the surplus generated by differentiated consumption. An individual firm takes $\mathcal{S}_t$ as given when setting its price, although aggregate shocks can move $\mathcal{S}_t$ across states and thereby affect marginal utility.

The consumer's first-order condition gives firm i the demand curve

$$y_{it} = \max \{a_t^{\text{eff}} + \xi_{it} - bp_{it}, 0\},$$

where

$$a_t^{\text{eff}} = a - \eta_Q Q_t, \quad Q_t = \int y_{it} di.$$

Each atomistic firm takes a_t^{eff} as given. Entry raises the mass of varieties and aggregate differentiated output, lowers the effective intercept, and toughens competition. Exit does the reverse.

Competition and the cross-section. The cross-sectional distribution affects the firm's problem through a single scalar. Integrating the demand curve across producing firms, and letting M_t denote their mass, $\Xi_t = \int \xi_{it} di$ the integral of their idiosyncratic demand shifters, and $P_t = \int p_{it} di$ the integral of their posted prices,

$$Q_t = M_t a_t^{\text{eff}} + M_t D_t + \Xi_t - bP_t.$$

Substituting the competition relation $a_t^{\text{eff}} = a - \eta_Q Q_t$ and solving for Q_t yields

$$Q_t = \frac{M_t a + M_t D_t + \Xi_t - bP_t}{1 + \eta_Q M_t},$$

and therefore

$$a_t^{\text{eff}} = A_{0t} - B_t D_t, \quad B_t = \frac{\eta_Q M_t}{1 + \eta_Q M_t}, \quad (12)$$

where $A_{0t} = (a - \eta_Q(\Xi_t - bP_t))/(1 + \eta_Q M_t)$.

Current idiosyncratic innovations average out across the continuum, while posted prices are determined jointly at the pricing stage. Each atomistic firm takes the aggregate price distribution as given. Thus D_t is the only aggregate demand innovation realized after prices are posted. Because prices are set before D_t is realized, the realized aggregate demand shock shifts the competition term firms face.

Production and markups. The outside sector produces the numeraire using a constant-returns technology $c_{0t} = A\ell_{0t}$. As long as the outside sector operates, its demand for the variable input is perfectly elastic and the equilibrium wage is $w_t = A$.

Each entrepreneur operates one differentiated firm. Firm i produces according to

$$y_{it} = z_{it} \ell_{it}^\nu, \quad \nu \in (0, 1],$$

where z_{it} is firm productivity and ℓ_{it} is the flexible input, hired after current shocks are realized. Its empirical counterpart is the combined labor-and-materials input used to construct markups in Section 4, and ν is disciplined by this input's share of total cost. Variable cost and marginal cost are therefore

$$C(y_{it}; z_{it}) = w_t \left(\frac{y_{it}}{z_{it}} \right)^{1/\nu}, \quad C_y(y_{it}; z_{it}) = \frac{w_t}{\nu} z_{it}^{-1/\nu} y_{it}^{(1-\nu)/\nu},$$

so marginal cost rises with output whenever $\nu < 1$.

Homogeneity of the variable-cost function gives $C_y y_{it} = C/\nu = w_t \ell_{it}/\nu$, so the markup is

$$\mu_{it} := \frac{p_{it}}{C_y(y_{it}; z_{it})} = \nu \frac{p_{it} y_{it}}{w_t \ell_{it}}.$$

The markup in the model is the output elasticity of its composite variable input divided by that input's revenue share. Its empirical counterpart is the cost-share markup of Section 4, constructed from combined labor and materials. Because ν is fixed by technology, the model markup varies only with the ratio of revenue to variable cost.

Firm i faces demand $y_{it} = \mathcal{D}(p_{it}; \xi_{it})$ and earns profits

$$\pi(p_{it}; \theta_{it}) = p_{it} y_{it} - w_t \left(\frac{y_{it}}{z_{it}} \right)^{1/\nu} - F,$$

where $F > 0$ is a per-period fixed operating cost.

Timing. Each period proceeds in three stages:

1. Entrepreneurs observe their lagged cost and demand states $\theta_{it-1} = (z_{it-1}, \xi_{it-1})$, the current industry uncertainty states $\sigma_{jt}^2 = (\sigma_{z,jt}^2, \sigma_{\xi,jt}^2)$, and the aggregate state $X_t = (Z_{t-1}, D_{t-1}, \sigma_{c,t}^2, \sigma_{d,t}^2)$. On this information, incumbents decide whether to continue operating or exit, and potential entrants decide whether to pay the sunk entry cost, drawing θ_0 from G if they do. Firms that operate then post prices p_{it} . An incumbent that exits does not produce this period.
2. Aggregate, industry, and firm-level cost and demand shocks are realized. Firms hire the labor required to produce the realized demand at the posted prices.
3. Agents consume.

The information structure is the same as in Sections 2 and 3. Firms observe the conditional distribution of shocks when choosing prices but not their current realizations and commit to satisfying demand at the posted prices. If firms could instead observe current shocks and freely reset prices before serving demand, the precautionary-pricing motive with respect to those shocks would disappear. More generally, the mechanism applies to uncertainty realized over the horizon over which firms are committed to their posted prices. The empirical markup responses discipline the quantitative importance of this restriction, while more frequent repricing or greater rationing flexibility would attenuate the mechanism. Appendix A microfound the posted-price commitment case.

Pricing and exit. Write

$$s_{it} = (\theta_{it-1}, \sigma_{jt}^2, X_t, B_t)$$

for the firm's state when exit and pricing decisions are made. The endogenous aggregate state B_t is a one-to-one transformation of the mass of active firms. The competition intercept is not an additional state: conditional on (X_t, B_t) , $A_{0t} = A_0(X_t, B_t)$ is determined contemporaneously in equilibrium.

A worker chooses at the beginning of each period whether to remain a worker or enter entrepreneurship. Let $W^w(X, B)$ denote the value of remaining a worker in the current period. Using the consumption-index representation above,

$$W^w(X, B) = \mathbb{E}[U(w\bar{\ell} + S) + \beta W(X', B') \mid X, B].$$

Let $J^e(X, B)$ denote the ex-ante value of entering, net of the entry cost but before drawing the initial firm state, and let $W(X, B)$ denote the value before the occupational choice. Then

$$W(X, B) = \max \{W^w(X, B), J^e(X, B)\}.$$

An incumbent that exits becomes a worker for the current period and may reconsider entry in subsequent periods. An incumbent that continues receives utility from wage income and realized firm profit together with differentiated-goods surplus, and retains the option to operate in the future. The incumbent's value function satisfies

$$V(s) = \max \left\{ \max_{p>0} \mathbb{E} \left[U(w\bar{\ell} + \mathcal{S} + \pi(p; \theta)) + \beta V(s') \mid s \right], W^w(X, B) \right\}. \quad (13)$$

Because each firm is atomistic, it takes differentiated-goods surplus as given. Moreover, its current price does not affect the transition law of the state. The pricing first-order condition is therefore

$$\mathbb{E} \left[U'(w\bar{\ell} + \mathcal{S} + \pi(p^*; \theta)) \cdot \pi_p(p^*; \theta) \mid s \right] = 0.$$

The continuation value does not enter the current price choice. Since an individual firm takes $\mathcal{S}$ as given, the pricing condition has the same form as in Section 3, with $w\bar{\ell} + \mathcal{S}$ replacing resources outside the firm. For industry and idiosyncratic uncertainty, $\mathcal{S}$ is unchanged because these shocks wash out in the aggregate, so the comparative statics of Section 3 apply directly: greater cost uncertainty raises the targeted markup, while greater demand uncertainty lowers it. Aggregate uncertainty can additionally move $\mathcal{S}$ and thereby alter marginal-utility weights; the quantitative counterfactuals incorporate this general-equilibrium channel.

Exit is an occupational decision. Risk aversion affects it because the continuation value depends on the distribution of future firm profits and differentiated-goods surplus, not only on their expectation.

Entry and free entry. Entry is the same occupational margin in reverse. A worker may become an entrepreneur by paying a sunk entry cost F_{entry} . Upon entry, the entrepreneur is assigned to an industry and draws its initial productivity and demand states from the stationary distribution G . Industry assignment determines the cost and demand uncertainty processes the firm subsequently faces. The value of entering with

initial state θ_0 is

$$V^0(\theta_0; \mathbf{X}_t, B_t) = \max_{p > 0} \mathbb{E} \left[U(w\bar{\ell} + \mathcal{S} + \pi(p; \theta; A_0(\mathbf{X}_t, B_t), B_t) - F_{\text{entry}}) + \beta V(s') \mid \theta_0, \mathbf{X}_t, B_t \right].$$

An entrant pays the entry cost, posts a price, and produces in the entry period; unlike an incumbent, it does not have an exit option before its first period of production. Before the initial firm state is drawn, the ex-ante value of entry is

$$J^e(\mathbf{X}_t, B_t) = \mathbb{E}_G[V^0(\theta_0; \mathbf{X}_t, B_t)].$$

Free entry requires

$$J^e(\mathbf{X}_t, B_t) \leq W^w(\mathbf{X}_t, B_t),$$

with equality whenever entry is positive.

The value of entry is increasing in the effective demand intercept because, under the linear demand system,

$$\frac{\partial \pi(p^*; \theta)}{\partial a_{\text{eff}}} = p^* - C_y > 0.$$

Free entry therefore determines the toughness of competition: when entrepreneurship becomes less attractive—for example, because profit risk rises—entry falls and the effective demand intercept rises until the free-entry condition is restored. The mass of entrants, the mass of operating firms, and the exit rate are therefore equilibrium outcomes.

Definition 2 (Recursive equilibrium). *Let Φ_t denote the cross-sectional distribution of active firms. A recursive competitive equilibrium consists of a wage, an effective-intercept function, firm value and pricing policies, an exit rule, entry and operating masses, the distribution Φ_t , and allocations such that: (i) households and firms optimize; (ii) the value of entry does not exceed the value of remaining a worker, with equality whenever entry is positive; (iii) the effective demand intercept is consistent with aggregate differentiated output; (iv) the cross-sectional distribution of active firms evolves according to the shock processes, firms' exit decisions, and the entry flow; and (v) all markets clear.*

Market clearing for each differentiated variety requires

$$c_{kt} = y_{kt}. \tag{14}$$

Let M_t^e denote the mass of entrants and ℓ_{0t} the labor used in the outside sector. Labor and numeraire market clearing require

$$\ell_{0t} + \int_{k \in \mathcal{M}_t} \ell_{kt} dk = \bar{\ell}, \quad (15)$$

$$\int_0^1 c_{0,it} di + FM_t + F_{\text{entry}}M_t^e = A\ell_{0t}. \quad (16)$$

The operating cost is paid by every active firm, whereas the entry cost is paid only by current entrants. Hence the aggregate entry resource cost is $F_{\text{entry}}M_t^e$.

For a given aggregate state and mass of active firms, the free-entry condition determines the contemporaneous effective demand intercept $A_{0t} = A_0(\mathbf{X}_t, B_t)$. Since

$$B_t = \frac{\eta_Q M_t}{1 + \eta_Q M_t},$$

B_t provides a one-to-one representation of the equilibrium mass of firms. Appendix E.2 describes the numerical solution, including the Krusell–Smith approximation and accuracy checks.

5.2 Estimation and Model Fit

We estimate four parameters by simulated method of moments: absolute risk aversion α , idiosyncratic productivity dispersion $\sigma_{z,\text{idio}}$, the demand slope b , and the operating cost F . The four targeted moments are the markup responses to cost and demand uncertainty, the manufacturing firm exit rate, and the aggregate markup. The shock and volatility processes, including the within-industry common-shock loadings κ_z and κ_ξ , are estimated directly from the empirical shock panel and the regressions in Section 4, prior to SMM estimation. The remaining shock-dispersion, common-shock, and sales-response moments are untargeted model diagnostics rather than additional SMM moments. Appendix E.4 describes the measurement and estimation steps, the assigned and measured parameters, and the model fit.

Table 3 reports the parameters assigned or measured before estimation, the four parameters estimated jointly by SMM, and the fit of the model to the targeted moments.

Table 3: Parameter values and model fit.

<i>Panel A. Externally estimated and assigned parameters</i>		
Parameter	Value	Source
Variable-input elasticity ν	0.90	Labor plus materials cost share
Cost-surprise SD	0.055	Section 4 shock panel
Demand-surprise SD	0.051	Section 4 shock panel
Industry persistence ρ_Z, ρ_ξ	0.49, 0.11	Section 4 shock panel
Aggregate persistence ρ_Z, ρ_D	0.56, 0.18	Aggregate shock processes
Aggregate variance shares s_Z, s_D	0.70, 0.66	Common year components
Common-shock loadings κ_Z, κ_ξ	0.77, 0.25	Within-industry covariance
<i>Panel B. Parameters estimated by SMM</i>		
Parameter	Estimate	Primary identifying moments
Risk aversion	2.20	Uncertainty coefficients
Idiosyncratic productivity SD	0.26	Exit rate
Demand slope	0.75	Relative uncertainty responses
Operating cost	0.06	Markup and exit rate
<i>Panel C. Targeted moments</i>		
Moment	Data	Model
Cost-uncertainty coefficient	0.010	0.006
Demand-uncertainty coefficient	-0.010	-0.006
Annual firm exit rate	0.072	0.072
Aggregate markup	1.30	1.30

Notes: The uncertainty coefficients are the IO-industry counterparts of the Section 4 IV estimates, matching the industry classification used in the simulated panel. The SMM criterion is 9.596, and the largest absolute discrepancy is 2.64 target standard deviations.

5.3 The Welfare Value of Precautionary Pricing

We ask a simple question, What are the welfare consequences of allowing firms to use prices to manage profit risk? To answer this question, we compare the estimated economy with precautionary pricing with one in which firms are forced to post the risk-neutral price. Importantly, preferences are identical in the two economies, and we solve each economy as a full free-entry equilibrium at the estimated parameters.

In the *estimated economy*, entrepreneurs bear profit risk and choose price optimally. The posted price solves

$$\mathbb{E}_{\theta|s} \left[U' \left(w\bar{\ell} + \mathcal{S}_t + \pi(p; \theta) \right) \pi_p(p; \theta) \right] = 0. \quad (17)$$

In the *risk-neutral-pricing economy*, entrepreneurs retain the same preferences and bear the same risk, but prices are constrained to maximize expected profits rather than

expected utility, that is, the risk-neutral price solves

$$\mathbb{E}_{\theta|s}[\pi_p(p; \theta)] = 0. \quad (18)$$

Welfare measurement. Each household pays R_t for differentiated goods and receives subutility $\mathcal{V}_t$. A worker's numeraire consumption is $c_{0t}^w = w\bar{\ell} - R_t$. An active entrepreneur i consumes

$$c_{0,it}^e = c_{0t}^w + \pi_{it} - \mathbf{1}_{\{i \text{ enters at } t\}} F_{\text{entry}},$$

where the operating cost is already included in profit. Utilitarian welfare is

$$\mathcal{W} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\int_{\mathcal{M}_t} U(c_{0,it}^e + \mathcal{V}_t) di + (1 - M_t) U(c_{0t}^w + \mathcal{V}_t) \right]. \quad (19)$$

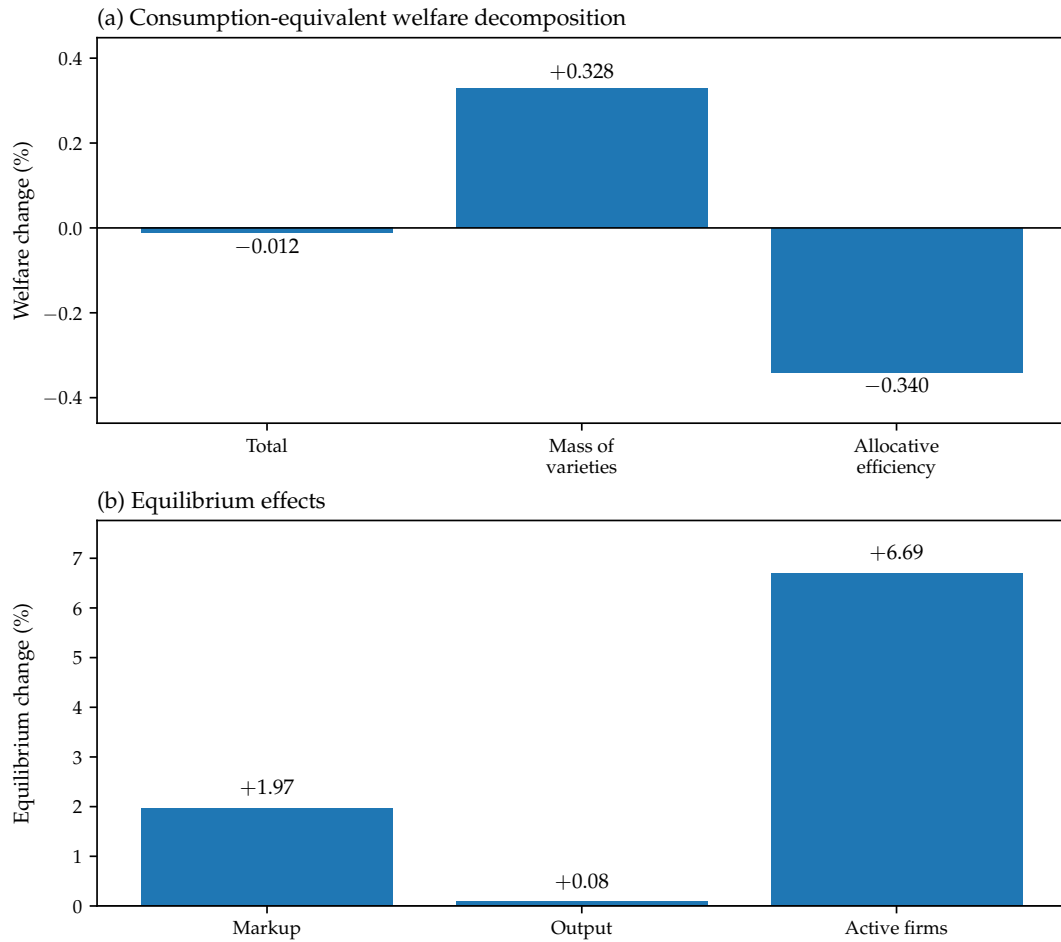
Thus every entrepreneur and worker enters the social welfare function with equal weight and the same preferences. Entry and operating costs enter exactly once through individual consumption and, equivalently, through the aggregate resource constraint.

We report welfare differences in consumption equivalent units. For economy j relative to economy r , $\lambda^{j,r}$ is the constant amount of numeraire consumption that must be given to every household in economy r , in every date and state, to make its utilitarian welfare equal to that in economy j . We report $100\lambda^{j,r}$ relative to per-household numeraire consumption in the risk-neutral-pricing economy. This calculation evaluates eq. (19) directly under the same preferences and involves no welfare approximation. Appendix E.3 gives the CARA formula used in the computation.

Figure 6 compares the estimated economy with the counterfactual in which firms maximize expected profits when setting prices.⁸ The central result is a tradeoff between product variety and allocative efficiency: precautionary markups reduce output and surplus per variety, but they also support a larger mass of firms and therefore more varieties. Quantitatively, precautionary pricing raises the aggregate markup by approximately 2 percent and the mass of active firms by 7 percent. The additional varieties contribute 0.33 percent of consumption to welfare, while lower allocative efficiency costs 0.34 percent, with the two effects almost perfectly offsetting each other.

⁸To locate the source of the welfare effect, we divide the equilibrium allocation into the mass of active firms and all remaining determinants of welfare, which include surplus per variety and the allocation of consumption and risk between workers and entrepreneurs. Starting from the risk-neutral-pricing economy, we replace these objects with their values in the estimated economy. We consider both possible orders of replacement and average each object's marginal contribution across the two orders. This gives an exact decomposition with no residual. We call the contribution of firm mass the mass-of-varieties channel and the contribution of the remaining allocation the allocative-efficiency channel.

Figure 6: Precautionary pricing vs. risk-neutral pricing.



Notes. Each bar reports the difference between the estimated economy with precautionary pricing and the risk-neutral-pricing counterfactual. Panel (a) decomposes the welfare effect into the mass-of-varieties and allocative-efficiency channels. Panel (b) reports the associated changes in equilibrium outcomes.

6 Conclusion

Why are markups cyclical, and why does their cyclicality vary across firms, industries, and business-cycle episodes? This paper argues that the answer depends on two forces: the realization of shocks and the composition of the risks firms face when prices are set. The first force is relatively mechanical. Realized cost and productivity shocks affect marginal cost and therefore move measured markups even when posted prices do not change; demand shocks may also affect realized marginal cost through scale effects—that is, by influencing the amount of output produced by firms. The second force is more subtle. When firms set prices before uncertainty is resolved, firms run by risk-averse owners or managers may use prices to manage their exposure to profit risk. Cost risk induces firms to raise prices and markups to reduce demand and thereby

limit exposure to high-cost states. Demand risk instead induces firms to lower prices and markups to sustain sales in weak-demand states. Measured markup cyclicalities therefore reflect both the effect of realized shocks on marginal cost and the way the cyclical composition of risk shapes firms' ex-ante pricing decisions. Because cost and demand risk vary over time and across firms, and affect prices in opposite directions, the cyclicalities of markups can differ across firms, industries, and business cycles.

We establish these results in a general pricing environment and test the central sign predictions of our theory using U.S. manufacturing data. Industries raise markups when uncertainty about input costs increases, and lower them when uncertainty about customer demand rises. The same asymmetry appears within firms, indicating that the industry results reflect changes in firms' pricing behavior rather than only changes in the composition of producers. We also find that markups of small and medium-sized firms respond more to uncertainty than those of larger and more diversified firms, consistent with the mechanism being stronger when owners bear more undiversified profit risk. We also decompose markup cyclicalities into the contributions of uncertainty and realized cost movements. We find that, from 1972 to 2018, the cyclical component of markups mostly reflects realized cost movements, especially from the 1990s onward. The contributions of uncertainty are larger in the earlier period, particularly in the 1970s and 1980s, and during recessions. In line with our understanding of the nature of previous business cycles, cost risk accounts for the vast majority of the uncertainty contributions to markups in the 1970s and 1980s, and demand risk largely explains the uncertainty effects around the early 1990s, the dot-com bubble, and the Great Recession.

To assess the quantitative importance and general-equilibrium consequences of the mechanism, we embed precautionary pricing in a stochastic general-equilibrium model with heterogeneous firms, entry and exit, and idiosyncratic and aggregate uncertainty. An estimated version of the model reproduces the opposing responses of markups to cost and demand uncertainty while jointly matching the average markup and the firm exit rate. The model implies that uncertainty effects account for only a modest share of measured markup cyclicalities because realized cost movements dominate the cyclical variation in ex-post markups, just as in the data. Yet the mechanism is quantitatively important for understanding aggregate outcomes. Preventing firms from adjusting prices in response to risk reduces entry and the equilibrium mass of varieties, and lowers aggregate welfare despite improving efficiency through lower markups.

Several limitations point to natural directions for future work. First, our empirical analysis likely understates the quantitative importance of precautionary pricing, particularly the role of idiosyncratic risk. In the NBER-CES data, aggregation to the industry level is likely to attenuate much of the firm-specific variation in cost and demand uncertainty, so that our estimates primarily capture the effects of aggregate risk. The

Compustat analysis preserves firm-level variation, but its sample consists of relatively large and diversified publicly traded firms, for which exposure to idiosyncratic risk may be more limited. Both features may therefore make the precautionary-pricing channel appear quantitatively smaller than it is for the broader population of firms. This matters not only for the magnitude of markup responses, but also for the share of markup cyclicalities that the mechanism can explain and for its implications for markup dispersion, misallocation, and allocative efficiency. Confidential U.S. Census micro-data would provide a natural complement to our analysis, allowing us to construct uncertainty and markup measures at a much finer level within manufacturing, separate aggregate from idiosyncratic risk, and assess how much of the precautionary-pricing mechanism is obscured by aggregation and sample selection.

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Appendices

“Precautionary Pricing and Markup Cyclicalities”

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A Microfoundations for Price Commitment

The baseline model's timing represents the interval between a seller's quote and fulfillment, rather than an indefinitely fixed price. This appendix provides a directed-search microfoundation for both price commitment and demand uncertainty. Sellers first use posted prices to attract buyers; buyer arrivals and input costs are then realized; and sellers honor the terms on which customers directed their search. Revising the price after observing arrivals would change the offer that attracted those customers, while rationing would amount to refusing transactions promised at that offer. Cost uncertainty, arising from input-price risk, is taken as primitive.

A.1 Environment

Time is discrete. In each period, a unit mass of buyers and a unit mass of sellers participate in a market for a homogeneous good. Each buyer demands at most one unit and has valuation v , drawn i.i.d. from a distribution F with support $[\underline{v}, \bar{v}]$ and continuous density f . Buyers observe their valuations prior to search.

Each seller produces output using one unit of an intermediate input per unit of output. The input is purchased in a competitive spot market at price

$$\bar{\kappa} + \eta,$$

where $\bar{\kappa} > 0$ and η is a mean-zero cost shock realized after output prices are posted. This captures environments in which firms commit to prices before upstream markets clear (e.g., catalog pricing prior to the realization of energy or commodity prices).

Sellers are risk-averse entrepreneurs with CARA preferences

$$U(\pi) = -\exp(-\alpha\pi),$$

where $\alpha > 0$.

The timing within period is as follows:

1. Sellers simultaneously post prices.
2. Buyers observe all posted prices and direct their search toward sellers.
3. Each buyer visits exactly one seller.
4. Input prices $\bar{\kappa} + \eta$ are realized.
5. Trade occurs at posted prices; sellers produce and deliver.

Thus, prices are set before sellers know either how many buyers will arrive or the realization of input costs.

A.2 Buyer Search and Demand

Consider a seller posting price p . In a symmetric directed-search equilibrium, buyers randomize across sellers posting the same price. Let $\lambda(p)$ denote the expected number of buyers visiting a seller that posts price p . Standard directed search arguments imply that $\lambda(p)$ is decreasing in p , reflecting the fact that higher prices attract fewer buyers. We assume

$$\lambda'(p) < 0.$$

Given a unit mass of buyers and independent search decisions, the realized number of arrivals n at a seller posting price p follows a Poisson distribution:

$$n \sim \text{Poisson}(\lambda(p)).$$

Each arriving buyer purchases one unit, so realized output equals arrivals:

$$y = n.$$

A.3 Seller Profits and Risk Decomposition

A seller posting price p earns profits

$$\pi(p; n, \eta) = (p - \bar{\kappa} - \eta)n.$$

Taking expectations over arrivals and cost shocks (which are independent),

$$\mathbb{E}[\pi(p)] = (p - \bar{\kappa})\lambda(p).$$

Writing realized demand as the sum of its mean and a deviation:

$$n = \lambda(p) + \xi, \quad \xi := n - \lambda(p),$$

where $\mathbb{E}[\xi] = 0$ and, under the Poisson specification,

$$\text{Var}(\xi) = \lambda(p).$$

Substituting yields

$$\begin{aligned}\pi(p; \xi, \eta) &= (p - \bar{\kappa} - \eta)[\lambda(p) + \xi] \\ &= \underbrace{(p - \bar{\kappa})\lambda(p)}_{\text{deterministic}} + \underbrace{(p - \bar{\kappa})\xi - \lambda(p)\eta - \eta\xi}_{\text{stochastic}}.\end{aligned}$$

This expression coincides with the profit representation in Section 2, with expected demand $y(p) = \lambda(p)$. Demand uncertainty arises endogenously from buyer arrival risk, while cost uncertainty originates from spot-market input prices.

If expected arrivals are linear in price,

$$\lambda(p) = a - bp, \quad a, b > 0,$$

then profits reduce to

$$\pi(p; \xi, \eta) = (p - \bar{\kappa})(a - bp + \xi) - (a - bp + \xi)\eta,$$

which matches exactly the profit function analyzed in Section 2 without demand-elasticity risk.

A.4 Optimal Pricing

The seller chooses p to maximize expected utility prior to observing ξ or η :

$$\max_{p>0} \mathbb{E}_{\xi, \eta} [-\exp(-\alpha \pi(p; \xi, \eta))].$$

This is precisely the pricing problem studied in Section 2. All those results carry over: cost uncertainty raises optimal prices, demand uncertainty lowers them, and the cyclicalities of markups depends on the relative importance of these two sources of risk.

Under the Poisson specification, demand variance satisfies $\sigma_\xi^2 = \lambda(p)$, so demand risk is increasing in expected sales. Evaluated at the risk-neutral optimum,

$$\sigma_\xi^2 = \lambda(p_{\text{RN}}) = a - bp_{\text{RN}} = \frac{a - b\bar{\kappa}}{2}.$$

Time variation in demand uncertainty can therefore arise from fluctuations in the mass of active buyers or in search intensity, while time variation in cost uncertainty reflects volatility in input markets.

B Omitted Proofs

B.1 Proof of Theorem 1

Theorem 1 (Optimal pricing under uncertainty). *Suppose that:*

- A1. **Price commitment.** *The firm sets p before θ is realized and commits to satisfying demand at that price, $y = \mathcal{D}(p; \theta)$. Thus output is determined by the posted price and realized demand, and is not subsequently adjusted in response to shocks.*
- A2. **Preferences.** *U is twice continuously differentiable, strictly increasing, and strictly concave.*
- A3. **Demand and technology.** *For almost all θ , $\mathcal{D}(\cdot; \theta)$ and $\mathcal{C}(\cdot; \theta)$ are twice continuously differentiable, with $\mathcal{D}_p(p; \theta) < 0$ and $\mathcal{C}_y(y; \theta) > 0$.*
- A4. **Interior optimum.** *The expected-utility problem admits a unique interior optimum.*
- A5. **Regularity.** *The expected-utility objective is finite and continuously differentiable in p in a neighborhood of the optimum, with its derivative obtained by differentiation under the expectation.*

Then the unique optimal price p^* satisfies

$$\mathbb{E}_\theta \left[U'(\mathcal{W}(p^*; \theta)) \pi_p(p^*; \theta) \right] = 0,$$

where

$$\pi_p(p; \theta) = y(p; \theta) + [p - \mathcal{MC}(p; \theta)] y_p(p; \theta).$$

Proof. Define the objective function $F(p) := \mathbb{E}_\theta[U(\omega(\theta) + \pi(p; \theta))]$. By A4, the expected-utility problem admits a unique interior optimum p^* . By A5, F is continuously differentiable in a neighborhood of p^* , with its derivative obtained by differentiation under the expectation. Therefore, the first-order condition at p^* is

$$F'(p^*) = \mathbb{E}_\theta \left[U'(\omega(\theta) + \pi(p^*; \theta)) \pi_p(p^*; \theta) \right] = 0.$$

Finally, by A3 and the chain rule,

$$\pi_p(p; \theta) = \mathcal{D}(p; \theta) + [p - \mathcal{C}_y(\mathcal{D}(p; \theta); \theta)] \mathcal{D}_p(p; \theta) = y(p; \theta) + [p - \mathcal{MC}(p; \theta)] y_p(p; \theta),$$

where $\mathcal{MC}(p; \theta) := \mathcal{C}_y(\mathcal{D}(p; \theta); \theta)$. □

B.2 Proof of Theorem 2

Theorem 2 (Local pricing formula around the risk-neutral price). *Suppose A1–A5 of Theorem 1 hold. Let the risk-neutral price p_{RN} be the unique interior maximizer of expected profits, characterized by*

$$\mathbb{E}_\theta[\pi_{p,\text{RN}}] = 0, \quad \mathbb{E}_\theta[\pi_{pp,\text{RN}}] < 0.$$

In addition, suppose:

A6. Local preference family. *Preferences belong to a one-parameter family $\{U_\alpha\}_{\alpha \geq 0}$ with $U_0(\mathcal{W}) = \mathcal{W}$ that nests risk neutrality at $\alpha = 0$, normalized so that $U'_0(\mathcal{W}) = 1$, and whose coefficient of absolute risk aversion satisfies*

$$A_\alpha(\mathcal{W}) := -\frac{U''(\mathcal{W})}{U'(\mathcal{W})} = \alpha A(\mathcal{W}),$$

where $A(\mathcal{W}) > 0$ is continuously differentiable on the relevant domain.

A7. Regularity. *The expected-utility objective is twice continuously differentiable in (p, α) in a neighborhood of $(p_{\text{RN}}, 0)$, with its derivatives obtained by differentiation under the expectation.*

Define the benchmark objects evaluated at p_{RN} as

$$\begin{aligned} \mathcal{W}_{\text{RN}}(\boldsymbol{\theta}) &:= \omega(\boldsymbol{\theta}) + \pi(p_{\text{RN}}; \boldsymbol{\theta}), & \pi_{\text{RN}}(\boldsymbol{\theta}) &:= \pi(p_{\text{RN}}; \boldsymbol{\theta}), \\ \pi_{p,\text{RN}}(\boldsymbol{\theta}) &:= \pi_p(p_{\text{RN}}; \boldsymbol{\theta}), & y_{p,\text{RN}}(\boldsymbol{\theta}) &:= y_p(p_{\text{RN}}; \boldsymbol{\theta}). \end{aligned}$$

For any arbitrary fixed reference level $\mathcal{W}_$, define*

$$G(\mathcal{W}) := \int_{\mathcal{W}_*}^{\mathcal{W}} A(s) ds,$$

and let $\bar{A} := \mathbb{E}_\theta[A(\mathcal{W}_{\text{RN}})]$. Then, for sufficiently small α , the optimal price $p(\alpha)$ satisfies

$$p(\alpha) = p_{\text{RN}} + \alpha \mathcal{R} + o(\alpha),$$

where the first-order risk adjustment admits the covariance representation

$$\mathcal{R} = \underbrace{\varphi \left[\frac{\bar{A} \cdot \text{Cov}_\theta(\mathcal{W}_{\text{RN}}, \pi_{p,\text{RN}})}{\mathbb{E}_\theta[y_{p,\text{RN}}]} \right]}_{\text{pure-risk motive}} + \underbrace{\frac{\text{Cov}_\theta(G(\mathcal{W}_{\text{RN}}) - \bar{A}\mathcal{W}_{\text{RN}}, \pi_{p,\text{RN}})}{\mathbb{E}_\theta[y_{p,\text{RN}}]}}_{\text{wealth effects}}.$$

Proof. For $\alpha \geq 0$, define the expected-utility objective

$$V(p, \alpha) := \mathbb{E}_\theta \left[U_\alpha \left(\omega(\theta) + \pi(p; \theta) \right) \right],$$

and let

$$F(p, \alpha) := \frac{\partial V(p, \alpha)}{\partial p}.$$

By A7, differentiation under the expectation is valid in a neighborhood of $(p_{\text{RN}}, 0)$, so

$$F(p, \alpha) = \mathbb{E}_\theta \left[U'_\alpha \left(\omega(\theta) + \pi(p; \theta) \right) \pi_p(p; \theta) \right].$$

The optimal price $p(\alpha)$ satisfies $F(p(\alpha), \alpha) = 0$.

By the normalization of A6, $U'_0(\mathcal{W}) = 1$. Hence, $F(p, 0) = \mathbb{E}_\theta [\pi_p(p; \theta)]$. By definition of p_{RN} ,

$$F(p_{\text{RN}}, 0) = \mathbb{E}_\theta [\pi_{p, \text{RN}}] = 0.$$

Differentiating F with respect to p and evaluating at $(p_{\text{RN}}, 0)$ gives

$$F_p(p_{\text{RN}}, 0) = \mathbb{E}_\theta [\pi_{pp, \text{RN}}] < 0.$$

Therefore, by the implicit function theorem, there exists a continuously differentiable optimal-price function $p(\alpha)$ in a neighborhood of $\alpha = 0$. Moreover,

$$p'(0) = -\frac{F_\alpha(p_{\text{RN}}, 0)}{F_p(p_{\text{RN}}, 0)}.$$

It remains to characterize $F_\alpha(p_{\text{RN}}, 0)$.

By A6,

$$-\frac{U''_\alpha(\mathcal{W})}{U'_\alpha(\mathcal{W})} = \alpha A(\mathcal{W}).$$

Equivalently,

$$\frac{\partial}{\partial \mathcal{W}} \log U'_\alpha(\mathcal{W}) = -\alpha A(\mathcal{W}).$$

Fix an arbitrary reference level $\mathcal{W}_*$. Integrating from $\mathcal{W}_*$ to $\mathcal{W}$ yields

$$\log U'_\alpha(\mathcal{W}) - \log U'_\alpha(\mathcal{W}_*) = -\alpha \int_{\mathcal{W}_*}^{\mathcal{W}} A(s) \, ds.$$

Using the definition

$$G(\mathcal{W}) := \int_{\mathcal{W}_*}^{\mathcal{W}} A(s) ds,$$

we obtain

$$U'_\alpha(\mathcal{W}) = c(\alpha) \exp[-\alpha G(\mathcal{W})],$$

where $c(\alpha) := U'_\alpha(\mathcal{W}_*)$.

Since $U'_0(\mathcal{W}) = 1$, $c(0) = 1$. Differentiating with respect to α and evaluating at $\alpha = 0$,

$$\left. \frac{\partial U'_\alpha(\mathcal{W})}{\partial \alpha} \right|_{\alpha=0} = c'(0) - G(\mathcal{W}).$$

Therefore,

$$F_\alpha(p_{\text{RN}}, 0) = \mathbb{E}_\theta \left[\left(c'(0) - G(\mathcal{W}_{\text{RN}}) \right) \pi_{p, \text{RN}} \right].$$

Using $\mathbb{E}_\theta[\pi_{p, \text{RN}}] = 0$, the term involving $c'(0)$ vanishes. Thus,

$$F_\alpha(p_{\text{RN}}, 0) = -\mathbb{E}_\theta \left[G(\mathcal{W}_{\text{RN}}) \pi_{p, \text{RN}} \right].$$

Since the mean of $\pi_{p, \text{RN}}$ is zero,

$$F_\alpha(p_{\text{RN}}, 0) = -\text{Cov}_\theta \left(G(\mathcal{W}_{\text{RN}}), \pi_{p, \text{RN}} \right).$$

It follows that

$$p'(0) = \frac{\text{Cov}_\theta \left(G(\mathcal{W}_{\text{RN}}), \pi_{p, \text{RN}} \right)}{\mathbb{E}_\theta[\pi_{pp, \text{RN}}]}.$$

Define $\mathcal{R} := p'(0)$. Since $p(\alpha)$ is twice continuously differentiable around $\alpha = 0$,

$$p(\alpha) = p_{\text{RN}} + \alpha \mathcal{R} + o(\alpha),$$

where

$$\mathcal{R} = \frac{\text{Cov}_\theta \left(G(\mathcal{W}_{\text{RN}}), \pi_{p, \text{RN}} \right)}{\mathbb{E}_\theta[\pi_{pp, \text{RN}}]}.$$

This establishes the local pricing formula.

Using the pass-through representation,

$$\mathcal{R} = \varphi \frac{\text{Cov}_{\theta}(G(\mathcal{W}_{\text{RN}}), \pi_{p,\text{RN}})}{\mathbb{E}_{\theta}[y_{p,\text{RN}}]}.$$

Define $\bar{A} := \mathbb{E}_{\theta}[A(\mathcal{W}_{\text{RN}})]$, and note that

$$G(\mathcal{W}_{\text{RN}}) = \bar{A}\mathcal{W}_{\text{RN}} + [G(\mathcal{W}_{\text{RN}}) - \bar{A}\mathcal{W}_{\text{RN}}].$$

By the linearity of covariance,

$$\text{Cov}_{\theta}(G(\mathcal{W}_{\text{RN}}), \pi_{p,\text{RN}}) = \bar{A}\text{Cov}_{\theta}(\mathcal{W}_{\text{RN}}, \pi_{p,\text{RN}}) + \text{Cov}_{\theta}(G(\mathcal{W}_{\text{RN}}) - \bar{A}\mathcal{W}_{\text{RN}}, \pi_{p,\text{RN}}).$$

Substitution into the pass-through representation yields the desired result. $\square$

B.3 Proof of Proposition 1

Proposition 1 (Decomposition of the pure-risk motive by source). *Let $\theta = (\eta, \xi, \psi)$ collect a marginal-cost shifter η , a demand-level shifter ξ , and a demand-elasticity shifter ψ . Suppose the assumptions of Theorem 2 hold. In addition, assume:*

A8. **Shock structure.** *The shocks have mean zero, are mutually independent, and have finite fourth moments.*

A9. **Local smoothness.** *For each p in a neighborhood of p_{RN} , the mappings $\theta \mapsto \pi(p; \theta)$ and $\theta \mapsto \omega(\theta)$ are sufficiently smooth in a neighborhood of $\theta = \mathbf{0}$.*

Let $\sigma_k^2 = \text{Var}(\theta_k)$, $\Sigma = \text{Var}(\theta)$, and, for $k, l \in \{\eta, \xi, \psi\}$, define the local exposures

$$\mathcal{W}_k := \left. \frac{\partial \mathcal{W}(p; \theta)}{\partial \theta_k} \right|_{(p, \theta) = (p_{\text{RN}}, \mathbf{0})} \quad \text{and} \quad \pi_{pk} := \left. \frac{\partial \pi_p(p; \theta)}{\partial \theta_k} \right|_{(p, \theta) = (p_{\text{RN}}, \mathbf{0})}.$$

Let $\mathcal{R}$ denote the risk adjustment from Theorem 2, and write

$$\mathcal{R} = \mathcal{R}_{\text{risk}} + \mathcal{R}_{\text{wealth}},$$

where $\mathcal{R}_{\text{risk}}$ denotes the pure-risk component and $\mathcal{R}_{\text{wealth}}$ the wealth-effects component.

Then the pure-risk component decomposes as

$$\mathcal{R}_{\text{risk}} = \sum_{k \in \{\eta, \xi, \psi\}} \mathcal{R}_{\text{risk}}^{(k)} + o(\|\Sigma\|),$$

where

$$\mathcal{R}_{\text{risk}}^{(k)} := \varphi \bar{A} \cdot \frac{\mathcal{W}_k \pi_{pk}}{\mathbb{E}_\theta[y_{p,\text{RN}}]} \sigma_k^2.$$

Proof. Fix $p = p_{\text{RN}}$. From Theorem 2, the pure-risk component of the risk adjustment is

$$\mathcal{R}_{\text{risk}} = \varphi \bar{A} \cdot \frac{\text{Cov}_\theta(\mathcal{W}_{\text{RN}}, \pi_{p,\text{RN}})}{\mathbb{E}_\theta[y_{p,\text{RN}}]}, \quad \bar{A} := \mathbb{E}_\theta[A(\mathcal{W}_{\text{RN}})].$$

To characterize the small-risk limit, write $\theta = \varepsilon \mathbf{z}$, where the distribution of $\mathbf{z}$ is fixed and satisfies A8, and let $\varepsilon \rightarrow 0$. By A9, a Taylor expansion around $\theta = \mathbf{0}$ gives

$$\begin{aligned} \mathcal{W}(p_{\text{RN}}; \varepsilon \mathbf{z}) &= \mathcal{W}(p_{\text{RN}}; \mathbf{0}) + \varepsilon \sum_k \mathcal{W}_k z_k + \mathcal{O}(\varepsilon^2 \|\mathbf{z}\|^2), \\ \pi_p(p_{\text{RN}}; \varepsilon \mathbf{z}) &= \pi_p(p_{\text{RN}}; \mathbf{0}) + \varepsilon \sum_k \pi_{pk} z_k + \mathcal{O}(\varepsilon^2 \|\mathbf{z}\|^2), \end{aligned}$$

where

$$\mathcal{W}_k = \left. \frac{\partial \mathcal{W}(p; \theta)}{\partial \theta_k} \right|_{(p, \theta) = (p_{\text{RN}}, \mathbf{0})} \quad \text{and} \quad \pi_{pk} = \left. \frac{\partial \pi_p(p; \theta)}{\partial \theta_k} \right|_{(p, \theta) = (p_{\text{RN}}, \mathbf{0})}.$$

Since $\mathbb{E}[\mathbf{z}] = \mathbf{0}$, centering these expressions yields

$$\begin{aligned} \mathcal{W}_{\text{RN}} - \mathbb{E}[\mathcal{W}_{\text{RN}}] &= \varepsilon \sum_k \mathcal{W}_k z_k + o(\varepsilon), \\ \pi_{p,\text{RN}} - \mathbb{E}[\pi_{p,\text{RN}}] &= \varepsilon \sum_k \pi_{pk} z_k + o(\varepsilon). \end{aligned}$$

Therefore,

$$\text{Cov}_\theta(\mathcal{W}_{\text{RN}}, \pi_{p,\text{RN}}) = \varepsilon^2 \sum_{k,l} \mathcal{W}_k \pi_{pl} \mathbb{E}[z_k z_l] + o(\varepsilon^2).$$

Mutual independence and mean zero imply $\mathbb{E}[z_k z_l] = 0$ for $k \neq l$. Hence,

$$\text{Cov}_\theta(\mathcal{W}_{\text{RN}}, \pi_{p,\text{RN}}) = \varepsilon^2 \sum_k \mathcal{W}_k \pi_{pk} \text{Var}(z_k) + o(\varepsilon^2).$$

Since $\sigma_k^2 = \text{Var}(\theta_k) = \varepsilon^2 \text{Var}(z_k)$, and $\|\Sigma\| = \mathcal{O}(\varepsilon^2)$, it follows that

$$\text{Cov}_\theta(\mathcal{W}_{\text{RN}}, \pi_{p,\text{RN}}) = \sum_k \mathcal{W}_k \pi_{pk} \sigma_k^2 + o(\|\Sigma\|).$$

Substituting into the expression for $\mathcal{R}_{\text{risk}}$ gives

$$\mathcal{R}_{\text{risk}} = \sum_k \varphi \bar{A} \cdot \frac{\mathcal{W}_k \pi_{pk}}{\mathbb{E}_{\theta}[y_{p,\text{RN}}]} \sigma_k^2 + o(\|\Sigma\|),$$

which establishes the result. $\square$

B.4 Proof of Proposition 2

Proposition 2 (Sufficient conditions to sign each source of risk). *Suppose the assumptions of Proposition 1 hold. Assume that, locally at $(p, \theta) = (p_{\text{RN}}, \mathbf{0})$, the firm-side exposures satisfy:*

$$\pi_{\eta} < 0, \quad \pi_{p\eta} > 0; \quad \pi_{\xi} > 0, \quad \pi_{p\xi} \geq 0; \quad \pi_{\psi} < 0, \quad \pi_{p\psi} < 0;$$

and that the corresponding total-resource exposures preserve the sign of the firm-profit exposures,

$$\mathcal{W}_{\eta} < 0, \quad \mathcal{W}_{\xi} > 0, \quad \mathcal{W}_{\psi} < 0.$$

Then:

1. Cost uncertainty raises prices: $\mathcal{R}_{\text{risk}}^{(\eta)} > 0$.
2. Demand-level uncertainty lowers prices: $\mathcal{R}_{\text{risk}}^{(\xi)} \leq 0$, with strict inequality if $\pi_{p\xi} > 0$.
3. Demand-elasticity uncertainty lowers prices: $\mathcal{R}_{\text{risk}}^{(\psi)} < 0$.

Proof. By Proposition 1, the shock-specific contribution of shock $k \in \{\eta, \xi, \psi\}$ to the pure-risk component is

$$\mathcal{R}_{\text{risk}}^{(k)} = \varphi \bar{A} \cdot \frac{\mathcal{W}_k \pi_{pk}}{\mathbb{E}_{\theta}[y_{p,\text{RN}}]} \sigma_k^2.$$

Since $\varphi > 0$, $\bar{A} > 0$, $\sigma_k^2 \geq 0$, and $\mathbb{E}_{\theta}[y_{p,\text{RN}}] < 0$, the sign of $\mathcal{R}_{\text{risk}}^{(k)}$ is the opposite of the sign of $\mathcal{W}_k \pi_{pk}$. For cost uncertainty, $\mathcal{W}_{\eta} < 0$ and $\pi_{p\eta} > 0$, so $\mathcal{R}_{\text{risk}}^{(\eta)} > 0$. For demand-level uncertainty, $\mathcal{W}_{\xi} > 0$ and $\pi_{p\xi} \geq 0$, and hence $\mathcal{R}_{\text{risk}}^{(\xi)} \leq 0$, with strict inequality whenever $\pi_{p\xi} > 0$. Finally, for demand-elasticity uncertainty, $\mathcal{W}_{\psi} < 0$ and $\pi_{p\psi} < 0$, so $\mathcal{R}_{\text{risk}}^{(\psi)} < 0$. $\square$

B.5 Proof of Proposition 4

Proposition 4 (Ex-ante aggregate markup cyclical). *Suppose the assumptions of Theorem 2 and Proposition 1 hold for every firm and period, and let the distribution of shocks vary over time through σ_{it}^2 . Suppose further that $\mathcal{R}_{it}$ is continuously differentiable in σ_{it}^2 in a*

neighborhood of a reference uncertainty vector. Holding firm and sector sales shares fixed at their reference values, aggregate ex-ante markup cyclicalities satisfies, to first order,

$$\text{Cov}(\mu_t^{\text{hsw,ea}}, Y_t) \approx \text{Cov}(\mu_{t,\text{RN}}^{\text{hsw,ea}}, Y_t) + \alpha \sum_{i \in \mathcal{I}_t} \frac{(\bar{\mu}^{\text{hsw,ea}})^2 \bar{\omega}_i \Lambda_i}{(\bar{\mu}_i^{\text{ea}})^2} \underbrace{\sum_{k \in \{\eta, \xi, \psi\}} \frac{\partial \mathcal{R}_{it}}{\partial \sigma_{k,it}^2}}_{\text{exposure channel}} \times \underbrace{\text{Cov}(\sigma_{k,it}^2, Y_t)}_{\text{cyclical channel}},$$

where all derivatives are evaluated at the reference uncertainty vector, and bars denote values at the reference allocation.

Proof. From the definition of ex-ante markup,

$$\mu_{it}^{\text{ea}}(\alpha) = \mu_i^{\text{ea}}(p_{it}(\alpha)).$$

A first-order expansion around the risk-neutral price gives

$$\mu_{it}^{\text{ea}}(\alpha) \approx \mu_{it,\text{RN}}^{\text{ea}} + \Lambda_i [p_{it}(\alpha) - p_{it,\text{RN}}].$$

Using the local pricing result $p_{it}(\alpha) \approx p_{it,\text{RN}} + \alpha \mathcal{R}_{it}$, we obtain

$$\mu_{it}^{\text{ea}}(\alpha) \approx \mu_{it,\text{RN}}^{\text{ea}} + \alpha \Lambda_i \mathcal{R}_{it}.$$

Holding firm sales shares fixed at their reference values, the economy-wide harmonic sales-weighted markup is

$$\mu_t^{\text{hsw,ea}} = \left(\sum_{i \in \mathcal{I}_t} \bar{\omega}_i (\mu_{it}^{\text{ea}})^{-1} \right)^{-1}.$$

The derivative of the aggregate markup with respect to firm i 's markup, evaluated at the reference allocation, is

$$\frac{\partial \mu_t^{\text{hsw,ea}}}{\partial \mu_i^{\text{ea}}} = \frac{(\bar{\mu}^{\text{hsw,ea}})^2 \bar{\omega}_i}{(\bar{\mu}_i^{\text{ea}})^2}.$$

A first-order expansion of the aggregate markup series therefore gives

$$\mu_t^{\text{hsw,ea}} \approx \mu_{t,\text{RN}}^{\text{hsw,ea}} + \alpha \sum_{i \in \mathcal{I}_t} \frac{(\bar{\mu}^{\text{hsw,ea}})^2 \bar{\omega}_i \Lambda_i}{(\bar{\mu}_i^{\text{ea}})^2} \mathcal{R}_{it}.$$

Taking covariances with aggregate activity,

$$\text{Cov}(\mu_t^{\text{hsw,ea}}, Y_t) \approx \text{Cov}(\mu_{it,\text{RN}}^{\text{hsw,ea}}, Y_t) + \alpha \sum_{i \in \mathcal{I}_t} \frac{(\bar{\mu}^{\text{hsw,ea}})^2 \bar{\omega}_i \Lambda_i}{(\bar{\mu}_i^{\text{ea}})^2} \text{Cov}(\mathcal{R}_{it}, Y_t).$$

Now expand the risk adjustment locally around a reference uncertainty vector $\bar{\sigma}_i^2$:

$$\mathcal{R}_{it} \approx \mathcal{R}_i(\bar{\sigma}_i^2) + \sum_{k \in \{\eta, \xi, \psi\}} \left. \frac{\partial \mathcal{R}_i}{\partial \sigma_k^2} \right|_{\bar{\sigma}_i^2} (\sigma_{k,it}^2 - \bar{\sigma}_{k,i}^2).$$

The reference terms are constant over time, so

$$\text{Cov}(\mathcal{R}_{it}, Y_t) \approx \sum_{k \in \{\eta, \xi, \psi\}} \frac{\partial \mathcal{R}_{it}}{\partial \sigma_{k,it}^2} \text{Cov}(\sigma_{k,it}^2, Y_t).$$

Substituting into the preceding expression establishes the result. $\square$

B.6 Proof of Proposition 5

Proposition 5 (Ex-post aggregate markup cyclicalilty). *Define the aggregate realized marginal-cost wedge as*

$$\mathcal{X}_t := \frac{\sum_{i \in \mathcal{I}_t} y_{it} (\text{mc}_{it} - \mathbb{E}_{\theta_{it}}[\text{mc}_{it}])}{\sum_{i \in \mathcal{I}_t} y_{it} \text{mc}_{it}}.$$

Then ex-post aggregate markup cyclicalilty admits the exact decomposition

$$\text{Cov}(\mu_t^{\text{hsw,ep}}, Y_t) = \underbrace{\text{Cov}(\mu_t^{\text{hsw,ea}}, Y_t)}_{\text{uncertainty channel}} - \underbrace{\text{Cov}(\mathcal{X}_t \mu_t^{\text{hsw,ea}}, Y_t)}_{\text{realized-cost channel}}.$$

Proof. By definition of the harmonic sales-weighted markup,

$$\mu_t^{\text{hsw,ep}} = \frac{\sum_{i \in \mathcal{I}_t} p_{it} y_{it}}{\sum_{i \in \mathcal{I}_t} y_{it} \text{mc}_{it}} \quad \text{and} \quad \mu_t^{\text{hsw,ea}} = \frac{\sum_{i \in \mathcal{I}_t} p_{it} y_{it}}{\sum_{i \in \mathcal{I}_t} y_{it} \mathbb{E}_{\theta_{it}}[\text{mc}_{it}]}.$$

It follows that

$$\frac{\mu_t^{\text{hsw,ep}}}{\mu_t^{\text{hsw,ea}}} = \frac{\sum_{i \in \mathcal{I}_t} y_{it} \mathbb{E}_{\theta_{it}}[\text{mc}_{it}]}{\sum_{i \in \mathcal{I}_t} y_{it} \text{mc}_{it}}.$$

Since

$$\sum_{i \in \mathcal{I}_t} y_{it} \mathbb{E}_{\theta_{it}}[\text{mc}_{it}] = \sum_{i \in \mathcal{I}_t} y_{it} \text{mc}_{it} - \sum_{i \in \mathcal{I}_t} y_{it} (\text{mc}_{it} - \mathbb{E}_{\theta_{it}}[\text{mc}_{it}]),$$

the definition of $\mathcal{X}_t$ implies

$$\mu_t^{\text{hsw, ep}} = (1 - \mathcal{X}_t) \mu_t^{\text{hsw, ea}}.$$

Taking covariances with Y_t gives

$$\text{Cov}(\mu_t^{\text{hsw, ep}}, Y_t) = \text{Cov}(\mu_t^{\text{hsw, ea}}, Y_t) - \text{Cov}(\mathcal{X}_t \mu_t^{\text{hsw, ea}}, Y_t).$$

□

B.7 Proof of Corollary 1

Corollary 1 (Risk-adjusted pricing formula). *Define the risk-adjusted expectation at p by*

$$\mathbb{E}_p^{\text{RA}}[X] := \frac{\mathbb{E}[U'(\omega(\theta) + \pi(p; \theta))X]}{\mathbb{E}[U'(\omega(\theta) + \pi(p; \theta))]}.$$

Under the assumptions of Theorem 1, the optimal price satisfies

$$\mathcal{MR}_{\text{RA}}(p^*) = \mathcal{MC}_{\text{RA}}(p^*),$$

where risk-adjusted marginal revenue and marginal cost are given by

$$\mathcal{MR}_{\text{RA}}(p) := p + \frac{\mathbb{E}_p^{\text{RA}}[y(p; \theta)]}{\mathbb{E}_p^{\text{RA}}[y_p(p; \theta)]}, \quad \mathcal{MC}_{\text{RA}}(p) := \frac{\mathbb{E}_p^{\text{RA}}[\mathcal{MC}(p; \theta) y_p(p; \theta)]}{\mathbb{E}_p^{\text{RA}}[y_p(p; \theta)]}.$$

Equivalently, these objects admit the decomposition

$$\begin{aligned} \mathcal{MR}_{\text{RA}}(p) &= \underbrace{p + \frac{\mathbb{E}[y(p; \theta)]}{\mathbb{E}[y_p(p; \theta)]}}_{\mathcal{MR}_{\text{mean}}(p)} + \underbrace{\left(\frac{\mathbb{E}_p^{\text{RA}}[y]}{\mathbb{E}_p^{\text{RA}}[y_p]} - \frac{\mathbb{E}[y]}{\mathbb{E}[y_p]} \right)}_{\mathcal{RA}_{\text{MR}}(p)} \\ \mathcal{MC}_{\text{RA}}(p) &= \underbrace{\frac{\mathbb{E}[\mathcal{MC}(p; \theta) y_p(p; \theta)]}{\mathbb{E}[y_p(p; \theta)]}}_{\mathcal{MC}_{\text{mean}}(p)} + \underbrace{\left(\frac{\mathbb{E}_p^{\text{RA}}[\mathcal{MC} y_p]}{\mathbb{E}_p^{\text{RA}}[y_p]} - \frac{\mathbb{E}[\mathcal{MC} y_p]}{\mathbb{E}[y_p]} \right)}_{\mathcal{RA}_{\text{MC}}(p)}. \end{aligned}$$

Proof. The first-order condition of Theorem 1 can be written as

$$\mathbb{E}_\theta[U'(\omega + \pi)y] + p^* \mathbb{E}_\theta[U'(\omega + \pi)y_p] - \mathbb{E}_\theta[U'(\omega + \pi)\mathcal{MC} y_p] = 0$$

using the identity $\pi_p(p; \theta) = y(p; \theta) + [p - \mathcal{MC}(p; \theta)]y_p(p; \theta)$, and all objects are evaluated at (p^*, θ) to simplify notation.

Since utility is strictly increasing by A2, marginal utility satisfies $U'(\omega + \pi) > 0$ almost surely, and hence $\mathbb{E}_\theta[U'(\omega(\theta) + \pi(p^*; \theta))] > 0$. Dividing the first-order condition by this term and using the definition of the risk-adjusted expectation yields

$$\mathbb{E}_{p^*}^{\text{RA}}[y] + p^* \mathbb{E}_{p^*}^{\text{RA}}[y_p] - \mathbb{E}_{p^*}^{\text{RA}}[\mathcal{MC} y_p] = 0.$$

By A3, $y_p(p; \theta) < 0$ for all $p > 0$ and almost all θ . Because the risk-adjusted expectation uses strictly positive weights, it follows that $\mathbb{E}_{p^*}^{\text{RA}}[y_p] < 0$. Dividing by this term and rearranging gives

$$p^* + \frac{\mathbb{E}_{p^*}^{\text{RA}}[y]}{\mathbb{E}_{p^*}^{\text{RA}}[y_p]} = \frac{\mathbb{E}_{p^*}^{\text{RA}}[\mathcal{MC} y_p]}{\mathbb{E}_{p^*}^{\text{RA}}[y_p]}.$$

By definition, the left-hand side is the risk-adjusted marginal revenue $\mathcal{MR}_{\text{RA}}(p^*)$ and the right-hand side is the risk-adjusted marginal cost $\mathcal{MC}_{\text{RA}}(p^*)$. The decompositions into mean components and risk adjustments follow by adding and subtracting the corresponding physical-probability expressions. $\square$

C Canonical Cases

This appendix specializes the general results to canonical monopolistic-pricing environments. We first consider CARA preferences with linear and isoelastic demand. These cases provide closed-form expressions for the source-specific pricing responses and illustrate how demand curvature shapes firms' exposure to cost and demand uncertainty. We then consider CRRA preferences, which introduce wealth-dependent risk aversion and provide a transparent role for the level of the entrepreneur's wealth. Throughout, we assume that non-profit resources are constant across states unless otherwise indicated, so that the entrepreneur bears the firm's profit risk entirely.

CARA utility and linear demand

Consider CARA preferences

$$U_\alpha(W) = -\exp(-\alpha W),$$

constant marginal cost

$$\text{mc}_{it} = \bar{\kappa}_i + \eta_{it},$$

and linear demand

$$y_{it} = a + \xi_{it} - (b + \psi_{it})p,$$

where $a, b > 0$, $a > b\bar{\kappa}$, and the shocks η, ξ, ψ are mutually independent, mean-zero shocks with finite fourth moments and time-varying variances $\sigma_{it}^2 = (\sigma_{\eta,it}^2, \sigma_{\xi,it}^2, \sigma_{\psi,it}^2)$.

Corollary 3 (CARA utility, linear demand, and constant marginal cost). *The risk-neutral price and markup are*

$$p_{i,RN} = \frac{a + b\bar{\kappa}_i}{2b}, \quad \mu_{i,RN} = \frac{p_{i,RN}}{\bar{\kappa}_i},$$

and cost pass-through is $\varphi = 1/2$. For sufficiently small α , the optimal price satisfies

$$p_{it}(\alpha) \approx p_{i,RN} + \alpha \mathcal{R}_{it},$$

where, to first order in uncertainty,

$$\mathcal{R}_{it} = \underbrace{\frac{a - b\bar{\kappa}_i}{4} \sigma_{\eta,it}^2}_{\text{pure cost risk}} - \underbrace{\frac{a - b\bar{\kappa}_i}{4b^2} \sigma_{\xi,it}^2}_{\text{pure demand-level risk}} - \underbrace{\frac{a(a + b\bar{\kappa}_i)(a - b\bar{\kappa}_i)}{8b^4} \sigma_{\psi,it}^2}_{\text{pure elasticity risk}}.$$

Because expected marginal cost equals $\bar{\kappa}_i$,

$$\mu_{it}^{ea}(\alpha) \approx \mu_{i,RN} + \frac{\alpha}{\bar{\kappa}_i} \mathcal{R}_{it}.$$

Hence, ex-ante aggregate markup cyclicalicity is

$$\begin{aligned} \text{Cov}(\mu_t^{hsw,ea}, Y_t) &\approx \alpha (\bar{\mu}^{hsw,ea})^2 \sum_{i \in \mathcal{I}_t} \frac{\bar{\omega}_i}{\bar{\kappa}_i (\bar{\mu}_i^{ea})^2} \\ &\times \left[\frac{a - b\bar{\kappa}_i}{4} \text{Cov}(\sigma_{\eta,it}^2, Y_t) - \frac{a - b\bar{\kappa}_i}{4b^2} \text{Cov}(\sigma_{\xi,it}^2, Y_t) - \frac{a(a + b\bar{\kappa}_i)(a - b\bar{\kappa}_i)}{8b^4} \text{Cov}(\sigma_{\psi,it}^2, Y_t) \right]. \end{aligned}$$

CARA utility and isoelastic demand

Consider the same preferences and marginal-cost process, but let demand be

$$y_{it} = (1 + \xi_{it}) p_{it}^{-(\varepsilon + \psi_{it})}, \quad \varepsilon > 1.$$

Define

$$\bar{p}_i = \frac{\varepsilon}{\varepsilon - 1} \bar{\kappa}_i, \quad L_i = \log \bar{p}_i.$$

Corollary 4 (CARA utility, isoelastic demand, and constant marginal cost). *The risk-neutral price is*

$$p_{it,RN} \approx \bar{p}_i + \frac{\bar{\kappa}_i L_i}{(\varepsilon - 1)^2} \sigma_{\psi,it}^2,$$

and therefore

$$\mu_{it}^{ea}(\alpha) \approx \mu_{it,RN} + \frac{\alpha}{\bar{\kappa}_i} \mathcal{R}_{it},$$

where the risk adjustment decomposes as

$$\mathcal{R}_{it}(\sigma_{it}^2) = \underbrace{C_{i\eta} \sigma_{\eta,it}^2}_{\text{pure cost risk}} - \underbrace{\Theta_{i\psi} \sigma_{\psi,it}^2}_{\text{pure elasticity risk}},$$

where the coefficients (evaluated at $\bar{p}_i$) are

$$C_{i\eta} := \frac{\left(\frac{\varepsilon-1}{\varepsilon}\right)^{\varepsilon-1}}{\bar{\kappa}_i^\varepsilon} \quad \text{and} \quad \Theta_{i\psi} := \frac{(\bar{p}_i - \bar{\kappa}_i)^2}{\varepsilon - 1} \bar{p}_i^{-\varepsilon} L_i.$$

Hence, ex-ante aggregate markup cyclicalities satisfies

$$\text{Cov}(\mu_t^{hsw,ea}, Y_t) \approx \text{Cov}(\mu_{t,RN}^{hsw,ea}, Y_t) + \alpha(\bar{\mu}^{hsw,ea})^2 \sum_{i \in \mathcal{I}_t} \frac{\bar{\omega}_i}{\bar{\kappa}_i(\bar{\mu}_i^{ea})^2} \left[C_{i\eta} \text{Cov}(\sigma_{\eta,it}^2, Y_t) - \Theta_{i\psi} \text{Cov}(\sigma_{\psi,it}^2, Y_t) \right].$$

Corollary 4 highlights the role of demand curvature in shaping precautionary pricing. As under linear demand, cost uncertainty raises the ex-ante markup, while elasticity uncertainty lowers it. By contrast, pure demand-level uncertainty has no first-order effect on the precautionary markup under isoelastic demand because multiplicative demand-level shocks scale profits proportionally across prices and therefore do not alter the marginal payoff from a price change at the risk-neutral optimum. Elasticity uncertainty also affects the risk-neutral price, so its contribution to aggregate markup cyclicalities operates through both the risk-neutral and precautionary components. Thus, procyclical cost uncertainty pushes aggregate ex-ante markups toward procyclicalities, whereas procyclical elasticity uncertainty pushes the precautionary component toward countercyclicalities.

CRRA utility and linear demand

Consider the same linear-demand and marginal-cost environment as in Corollary 3, but let preferences be CRRA,

$$U_\alpha(\mathcal{W}) = \frac{\mathcal{W}^{1-\alpha}}{1-\alpha}, \quad \mathcal{W}_{it} > 0,$$

with the logarithmic case defined by continuity. Absolute risk aversion is

$$A_\alpha(\mathcal{W}) = \frac{\alpha}{\mathcal{W}},$$

so in the notation of Theorem 2,

$$A(\mathcal{W}) = \frac{1}{\mathcal{W}}, \quad G(\mathcal{W}) = \log \mathcal{W}$$

up an irrelevant additive constant.

Corollary 5 (CRRA utility, linear demand, and constant marginal cost). *The ex-ante markup satisfies*

$$\mu_{it}^{ea}(\alpha) \approx \mu_{i,RN} + \frac{\alpha}{\bar{\kappa}_i} \mathcal{R}_{it}^{CRRA},$$

where

$$\mathcal{R}_{it}^{CRRA} = \bar{A}_{it} \mathcal{R}_{it}^{CARA} + \mathcal{R}_{it}^{wealth},$$

with

$$\bar{A}_{it} = \mathbb{E} \left[\frac{1}{\mathcal{W}_{it,RN}} \right] \quad \text{and} \quad \mathcal{R}_{it}^{wealth} = \frac{1}{2} \frac{\text{Cov}(\log \mathcal{W}_{it,RN} - \bar{A}_{it} \mathcal{W}_{it,RN}, \pi_{p,it,RN})}{\mathbb{E}[y_{p,it,RN}]}.$$

Hence, ex-ante aggregate markup cyclicalities satisfies

$$\text{Cov}(\mu_t^{hsw,ea}, Y_t) \approx \alpha (\bar{\mu}^{hsw,ea})^2 \sum_{i \in \mathcal{I}_t} \frac{\bar{\omega}_i}{\bar{\kappa}_i (\bar{\mu}_i^{ea})^2} \left[\text{Cov}(\bar{A}_{it} \mathcal{R}_{it}^{CARA}, Y_t) + \text{Cov}(\mathcal{R}_{it}^{wealth}, Y_t) \right].$$

Corollary 5 nests the CARA linear-demand case while introducing two effects of wealth-dependent risk aversion. The pure-risk component preserves the CARA pricing response but scales it by average absolute risk aversion, $\bar{A}_{it}$. Thus, unlike under CARA preferences, the strength of the pricing response depends on the level of the entrepreneur's wealth: a given exposure to firm-level risk generates a smaller precautionary response when total wealth is larger. In addition, the wealth-effects term captures variation across states in the value placed on the redistribution generated by a price change and may reinforce or attenuate the pure-risk response.

CRRA utility and isoelastic demand

Consider the same isoelastic-demand and marginal-cost environment as in Corollary 4, but let preferences be CRRA,

$$U_\alpha(\mathcal{W}) = \frac{\mathcal{W}^{1-\alpha}}{1-\alpha}, \quad \mathcal{W}_{it} > 0,$$

with the logarithmic case defined by continuity. As above,

$$A(\mathcal{W}) = \frac{1}{\mathcal{W}}, \quad G(\mathcal{W}) = \log \mathcal{W}$$

up an irrelevant additive constant.

Corollary 6 (CRRA utility, isoelastic demand, and constant marginal cost). *The risk-neutral price is*

$$p_{it,RN} \approx \bar{p}_i + \frac{\bar{\kappa}_i L_i}{(\varepsilon - 1)^2} \sigma_{\psi,it}^2.$$

The ex-ante markup satisfies

$$\mu_{it}^{ea}(\alpha) \approx \mu_{it,RN} + \frac{\alpha}{\bar{\kappa}_i} \mathcal{R}_{it}^{CRRA},$$

where

$$\mathcal{R}_{it}^{CRRA} = \bar{A}_{it} \mathcal{R}_{it}^{CARA} + \mathcal{R}_{it}^{wealth},$$

with

$$\bar{A}_{it} = \mathbb{E} \left[\frac{1}{\mathcal{W}_{it,RN}} \right] \quad \text{and} \quad \mathcal{R}_{it}^{wealth} = \varphi_{it} \frac{\text{Cov}(\log \mathcal{W}_{it,RN} - \bar{A}_{it} \mathcal{W}_{it,RN}, \pi_{p,it,RN})}{\mathbb{E}[y_{p,it,RN}]}.$$

Hence, ex-ante aggregate markup cyclicalities satisfies

$$\begin{aligned} \text{Cov}(\mu_t^{hsw,ea}, Y_t) &\approx \text{Cov}(\mu_{t,RN}^{hsw,ea}, Y_t) \\ &+ \alpha (\bar{\mu}^{hsw,ea})^2 \sum_{i \in \mathcal{I}_t} \frac{\bar{\omega}_i}{\bar{\kappa}_i (\bar{\mu}_i^{ea})^2} \left[\text{Cov}(\bar{A}_{it} \mathcal{R}_{it}^{CARA}, Y_t) + \text{Cov}(\mathcal{R}_{it}^{wealth}, Y_t) \right]. \end{aligned}$$

Corollary 6 nests the CARA isoelastic-demand case while introducing the same two effects of wealth-dependent risk aversion as in the linear-demand environment. The pure-risk component preserves the corresponding CARA pricing response but scales it by average absolute risk aversion, $\bar{A}_{it}$, while the wealth-effects component may reinforce or attenuate that response.

D Robustness of the Markup Response

The baseline results establish that cost uncertainty raises markups while demand uncertainty lowers them. This section examines the robustness of these findings along several dimensions. At the industry level, we consider alternative measures of uncertainty, alternative samples, alternative markup measures, a longer window for constructing uncertainty, and specifications with year fixed effects. We then examine the robustness of the results in the firm-level specification using alternative inference procedures and measures of markups and variable costs.

Industry-level robustness. Tables 4 and 5 report robustness exercises for the industry-level IV specification, based on the NBER-CES manufacturing sample (1972–2018). Table 4 varies the measure of uncertainty and the estimation sample. The baseline results are similar when uncertainty is measured using downside semideviations and when the oil-shock or financial-crisis years are excluded. The estimated responses are particularly strong in the high-materials-share subsample, for which the flexible-input markup is measured most cleanly.

Table 4: Robustness to alternative uncertainty measures and samples.

Dependent variable: log markup	(1) Baseline	(2) Downside	(3) No 1973–75	(4) No 2008–10	(5) High-materials
Cost uncertainty $\widehat{\sigma}_{C,jt}$	+1.3* (+1.9)	+1.3* (+1.8)	+1.6*** (+2.9)	+1.6** (+2.5)	+1.5*** (+3.1)
Demand uncertainty $\widehat{\sigma}_{D,jt}$	−1.2** (−2.1)	−1.4* (−2.0)	−1.5** (−2.4)	−1.0* (−1.6)	−1.7*** (−3.0)
First-moment controls (cost, demand)	✓	✓	✓	✓	✓
First moments instrumented	—	—	—	—	—
Industry fixed effects	✓	✓	✓	✓	✓
Year fixed effects	—	—	—	—	—
First-stage F (cost / demand)	53/440	62/140	150/746	45/468	55/314
Observations	16,751	16,751	12,136	15,677	8,341

Notes. The table reports robustness exercises for the baseline IV specification in column (2) of Table 1, using NBER-CES manufacturing data from 1972–2018. Column (1) reproduces the baseline specification. Column (2) replaces the symmetric volatility measures with downside semideviations. Column (3) excludes 1973–1975, column (4) excludes 2008–2010, and column (5) restricts the sample to industries with high materials shares. Coefficients report the response to a one-standard-deviation increase in uncertainty, in log points (coefficient $\times 100$). Parentheses report t -statistics based on two-way clustering by industry and year. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table 5 considers alternative markup measures, a longer window for constructing uncertainty, and the inclusion of year fixed effects. The labor-only markup produces the same qualitative patterns as the baseline. The seven-year uncertainty window also yields estimates with the baseline signs, though they are less precise. Adding year fixed

effects absorbs common time variation and therefore identifies the relationship from within-year cross-industry variation, so this specification addresses a different source of identifying variation from the baseline.

Table 5: Robustness to alternative markup measures, uncertainty windows, time effects.

Dependent variable: log markup	(1) Baseline	(2) Labor-only	(3) Materials-only	(4) Seven-year window	(5) Year effects
Cost uncertainty $\widehat{\sigma}_{C,jt}$	+1.3* (+1.9)	+7.8*** (+2.8)	-0.8* (-1.9)	+0.9 (+1.0)	+0.8 (+0.8)
Demand uncertainty $\widehat{\sigma}_{D,jt}$	-1.2** (-2.1)	-5.4** (-2.4)	+0.2 (+0.4)	-1.4 (-1.6)	+1.8*** (+3.6)
First-moment controls (cost, demand)	✓	✓	✓	✓	✓
First moments instrumented	—	—	—	—	—
Industry fixed effects	✓	✓	✓	✓	✓
Year fixed effects	—	—	—	—	✓
First-stage F (cost / demand)	53/440	53/440	53/440	21/127	107/330
Observations	16,751	16,751	16,751	16,020	16,751

Notes. This table reports robustness exercises for the baseline specification in column (2) of Table 1. Column (1) reproduces the baseline specification. Columns (2) and (3) use the labor-only and materials-only markup measures, respectively. Column (4) constructs uncertainty using the standard deviation of seven rather than five trailing surprises. Column (5) adds year fixed effects, thereby absorbing common time variation and identifying the relationship from within-year cross-industry variation. Reporting conventions are those of Table 4.

Firm-level robustness. Table 6 examines the robustness of the within-firm results based on the Compustat sample (1972–2018). The first specification uses two-way clustering by firm and year, the second uses the unwinsorized markup measure, and the third constructs the markup using a broader variable-cost bundle that adds selling, general and administrative expenses (SG&A) to the costs of goods sold (COGS). The estimated coefficients preserve the predicted signs across all three specifications, but statistical significance varies across specifications. The negative response to demand uncertainty is significant under two-way clustering and the unwinsorized markup, while the positive response to cost uncertainty is strongest and statistically significant when the broader variable-cost bundle is used.

Table 6: Firm-level robustness, Compustat manufacturing sample (1972–2018).

Dependent variable: log markup	Cost uncertainty	Demand uncertainty	Obs.
Two-way clustered (firm and year)	+0.7 (+1.1)	-1.0* (-1.8)	102,689
Unwinsorized	+0.8 (+1.5)	-1.1** (-2.2)	102,689
COGS + SG&A cost bundle	+1.1** (+2.3)	-0.6 (-1.3)	102,689

E Quantitative Model Appendix

E.1 Savings and the Strength of Precautionary Pricing

The quantitative model of Section 5 shuts the bond market, so pricing is the only risk-management instrument. This appendix shows that under the model's CARA and quasilinear structure, an interior savings margin attenuates the quantitative strength of precautionary pricing without changing its qualitative predictions. Savings reduce the effective risk aversion relevant for pricing, so the estimated α can be interpreted as effective risk aversion net of self-insurance. Let $\mathcal{W} = a + w + \pi$ denote total resources including bond holdings a at price $q \in (0, 1)$, $\hat{V}$ the post-realization value, and $\tilde{\beta}$ the effective discount factor.

Proposition 6 (Effective risk aversion with savings). *Consider preferences (11) with CARA period utility $U(x) = -\exp(-\alpha x)$, $\alpha > 0$, over the consumption index $x = c_0 + \mathcal{V}(\{c_k\})$. Consider the inner problem in steady state with constant aggregate prices and a bond price $q \in (0, 1)$, and assume the borrowing constraint is slack. Then the inner value function inherits the CARA form in total resources:*

$$\hat{V}(\mathcal{W}; \theta) = -B(\theta) \exp(-\alpha_{\text{eff}} \mathcal{W}),$$

with $B(\theta) > 0$, and effective absolute risk aversion with respect to total resources is

$$\alpha_{\text{eff}} = \alpha(1 - q) < \alpha.$$

Proof. Quasilinearity implies that differentiated-goods consumption is independent of wealth. Define net differentiated-goods surplus by

$$\mathcal{S}(\theta) := \max_{\{c_k\}} \left\{ \mathcal{V}(\{c_k\}; \theta) - \int p_k c_k dk \right\}.$$

After optimizing over differentiated-goods consumption, the consumption index period utility is therefore

$$x = \mathcal{W} + \mathcal{S}(\theta) - q a'$$

Conjecture

$$\hat{V}(\mathcal{W}; \theta) = -B(\theta) \exp(-\alpha_{\text{eff}} \mathcal{W}), \quad V(a; \theta_{-1}) = -D(\theta_{-1}) \exp(-\alpha_{\text{eff}} a),$$

with $B, D > 0$ to be determined.

The inner problem becomes

$$\hat{V}(\mathcal{W}; \theta) = \max_{a'} \left\{ -\exp(-\alpha(\mathcal{W} + \mathcal{S}(\theta) - qa')) + \mathcal{V}^*(\theta) - \tilde{\beta} \tilde{D}(\theta) \exp(-\alpha_{\text{eff}} a') \right\},$$

where $\tilde{D}(\theta) := \mathbb{E}[D(\theta') | \theta]$.

The first-order condition for a' is

$$\alpha q \exp(-\alpha(\mathcal{W} + \mathcal{S}(\theta) - qa'^*)) = \tilde{\beta} \alpha_{\text{eff}} \tilde{D}(\theta) \exp(-\alpha_{\text{eff}} a'^*).$$

Taking logarithms and rearranging gives a savings policy that is linear in $\mathcal{W}$,

$$a'^*(\mathcal{W}; \theta) = \frac{\alpha}{\alpha_{\text{eff}} + \alpha q} (\mathcal{W} + \mathcal{S}(\theta)) + \frac{1}{\alpha_{\text{eff}} + \alpha q} \log \left(\frac{\tilde{\beta} \alpha_{\text{eff}} \tilde{D}(\theta)}{\alpha q} \right).$$

The optimized consumption index is therefore

$$x^*(\mathcal{W}; \theta) = \frac{\alpha_{\text{eff}}}{\alpha_{\text{eff}} + \alpha q} (\mathcal{W} + \mathcal{S}(\theta)) - \frac{q}{\alpha_{\text{eff}} + \alpha q} \log \left(\frac{\tilde{\beta} \alpha_{\text{eff}} \tilde{D}(\theta)}{\alpha q} \right).$$

Applying the envelope theorem gives

$$\hat{V}_{\mathcal{W}}(\mathcal{W}; \theta) = U'(x^*) = \alpha \exp(-\alpha x^*).$$

Matching the dependence on $\mathcal{W}$ to the derivative of the conjectured value function, $\alpha_{\text{eff}} B(\theta) \exp(-\alpha_{\text{eff}} \mathcal{W})$, requires $\alpha_{\text{eff}} = \alpha(1 - q)$.

Substituting the optimal policy back into the inner problem determines a positive state-dependent coefficient $B(\theta)$. Since total resources are affine in bond holdings,

$$\mathcal{W} = a + w + \pi(p^*; \theta),$$

the conjectured value function factorizes into a state-dependent term and $\exp(-\alpha_{\text{eff}} a)$. Taking expectations over $\theta | \theta_{-1}$ yields

$$V(a; \theta_{-1}) = -D(\theta_{-1}) \exp(-\alpha_{\text{eff}} a)$$

for some $D(\theta_{-1}) > 0$, verifying the conjectured functional form. The condition $q \in (0, 1)$ ensures $\alpha_{\text{eff}} \in (0, \alpha)$. $\square$

Interpretation. Without savings ($q = 0$), effective risk aversion with respect to total resources coincides with the primitive α . With savings, effective risk aversion $\alpha_{\text{eff}} = \alpha(1 - q)$ is lower because part of a wealth shock is absorbed through the savings margin

rather than current consumption. Combined with Theorem 1 and the CARA closed forms of Corollary 3, this implies that the static risk adjustment rescales to

$$\mathcal{R}^{\text{dynamic}} = (1 - q) \mathcal{R}^{\text{static}}.$$

An interior savings margin therefore attenuates the quantitative strength of precautionary pricing through lower effective risk aversion while preserving its qualitative sign predictions. The estimated risk aversion in Section 5.2 can accordingly be interpreted as effective risk aversion net of self-insurance.

Extension to time-varying q_t . Along a MIT transition path with a time-varying sequence $\{q_t\}$, the conjecture $\hat{V}_t(\mathcal{W}; \theta) = -B_t(\theta) \exp(-\alpha_{\text{eff},t} \mathcal{W})$ is verified period by period with $\alpha_{\text{eff},t} = \alpha(1 - q_t)$, so the dynamic risk-adjustment relation $\mathcal{R}_t^{\text{dynamic}} = (1 - q_t) \mathcal{R}_t^{\text{static}}$ holds at every date.

E.2 Model Solution

We solve the model by a Krusell–Smith algorithm in which the cross-sectional distribution enters the firm’s problem through B , the one-to-one transformation of the mass of active firms defined in (12). Firm productivity, demand, and industry uncertainty are represented by finite-state Markov chains. Aggregate uncertainty is also represented by a finite-state chain; innovations to aggregate cost and demand levels are integrated by Gaussian quadrature, with interpolation between aggregate-state nodes. On a finite grid for B , we conjecture

$$B_{t+1} = c(\mathbf{X}_t, \mathbf{X}_{t+1}) + \rho_B B_t,$$

where c is linear in the current and next aggregate states. The rule is defined for every connected pair of aggregate nodes and linearly interpolated across the B grid.

The surplus S_t is a second equilibrium object the firm’s state does not carry: it is a functional of the cross-sectional distribution, realized at the period’s aggregate demand draw. Firms evaluate it through a second fitted function, of the same class as the forecasting rule. We project the simulated S_t on a quadratic in the aggregate state, B_t , and the realized aggregate demand level—an intercept, B_t , B_t^2 , the four components of $\mathbf{X}_t$, D_t , D_t^2 , $B_t D_t$, and $D_{t-1} D_t$ —and evaluate the fitted law at every aggregate quadrature draw. Under CARA the fitted surplus multiplies every agent’s flow utility by $e^{-\alpha \hat{S}}$, which is how it enters the pricing objective, the value function, and the entry and worker values below.

For a conjectured rule, we solve the firm's problem and free entry jointly. At every (X, B) node, free entry determines A_0 by equating the expected entrant and worker values. We solve these conditions by a safeguarded Broyden method, warm-starting the value function and Jacobian. Given (A_0, B) , the firm chooses its price by maximizing expected current utility, weighted by $e^{-\alpha \widehat{S}}$ at each aggregate draw, over aggregate quadrature nodes and firm-level transitions, respecting the zero-demand boundary. Because the current price does not affect the transition law, the continuation value does not enter this maximization. We solve for the value net of the worker's, $\hat{V}(s) = V(s) - W(X, B)$: under CARA the worker's flow at each aggregate draw carries the same surplus weight, so the netted flow is the weighted expectation of $U(w\bar{\ell} + \pi) - U(w\bar{\ell})$ and the exit branch of the netted Bellman is zero. Given the resulting flow payoff, we iterate

$$\hat{V}(s) = \max\{\text{flow}(s) + \beta \mathbb{E}[\hat{V}(s') | s], 0\}$$

using safeguarded Anderson acceleration. Comparing the continuation value with the worker value yields the exit threshold at every state.

We then simulate a fixed aggregate history while evolving the cross-sectional measure without sampling error. Each date, we apply exit to incumbents, choose entry to satisfy the competition identity, compute aggregates, and advance the surviving measure with the Markov matrices. After a burn-in, we regress B_{t+1} on B_t and the current and next aggregate states, refit the surplus projection on the simulated path, damp the update of the forecasting rule, rebuild the B grid around its simulated range, and repeat until both fitted objects converge.

The final pass uses the reported Bellman and free-entry tolerances. We require both residuals to satisfy their tolerances, the simulated path to remain inside the B grid, and the refitted rule and refitted surplus projection to remain within the convergence tolerance of the versions used to generate the simulation. We report the forecasting rule's one-step R^2 and the simulation-based forecast error proposed by [Den Haan \(2010\)](#), obtained by iterating the forecasting rule along the full history without feeding the realized B_t back into the forecast, and the surplus projection's R^2 and maximum absolute error along the simulated path.

E.3 Welfare Computation

For CARA utility, define the stationary population sum of exponential utility kernels in economy j as

$$S^j = \mathbb{E} \left[\int_{\mathcal{M}_t^j} e^{-\alpha(c_{0,it}^{e,j} + \mathcal{V}_t^j)} di + (1 - M_t^j) e^{-\alpha(c_{0,t}^{w,j} + \mathcal{V}_t^j)} \right].$$

Stationarity implies $\mathcal{W}^j = -S^j/[\alpha(1 - \beta)]$. Adding λ units of numeraire consumption to every household in the reference economy multiplies S^r by $e^{-\alpha\lambda}$. Equating the resulting welfare to $\mathcal{W}^j$ therefore gives the exact consumption equivalent

$$\lambda^{j,r} = \frac{1}{\alpha} \log \left(\frac{S^r}{S^j} \right).$$

We calculate S^j and S^r over the same stationary aggregate history. Individual consumption includes profits and entry costs as specified in Section 5, so no additional welfare adjustment is applied.

E.4 Estimation

Simulated Method of Moments (SMM). We jointly estimate four structural parameters $\vartheta = (\alpha, \sigma_{z,\text{idio}}, b, F)$: absolute risk aversion, the time-invariant dispersion of firm productivity, the demand slope, and the operating cost. The four targeted moments are the two markup–uncertainty coefficients of Section 4, the manufacturing firm exit rate, and the aggregate markup. The empirical slopes are 1.3 log points for cost uncertainty and -1.2 log points for demand uncertainty. The manufacturing exit rate is 0.072, and the harmonic sales-weighted markup is 1.30. The within-industry common-shock loadings κ_z and κ_ξ , along with the other shock-process parameters, are estimated outside the SMM and held fixed during estimation. The estimator minimizes

$$Q(\vartheta) = \left(\mathbf{m}(\vartheta) - \mathbf{m}^{\text{data}} \right)' \Omega^{-1} \left(\mathbf{m}(\vartheta) - \mathbf{m}^{\text{data}} \right),$$

where $\mathbf{m}(\vartheta)$ is the four-element vector of simulated target moments and Ω is their data-moment covariance matrix. Its only off-diagonal entry is the 2×2 block for the two uncertainty coefficients, which uses the shift-share IV standard errors and their estimated correlation of -0.52 . The exit-rate and aggregate-markup diagonal entries are declared target tolerances because their sampling covariance with the regression coefficients is not estimated. The weighting matrix is fixed before optimization.

Each evaluation of $\mathbf{m}(\vartheta)$ solves the full free-entry equilibrium and simulates an industry panel. We then apply to the simulated data the same measurement and

econometric procedure used in Section 4, with one exception noted below. Specifically, we simulate 230 industries, corresponding to the input-output units at which the empirical shocks are constructed, with 50 heterogeneous firms per industry over 47 years. Each industry draws paths for cost and demand volatility from the estimated stochastic-volatility processes.

We aggregate firm outcomes into harmonic sales-weighted industry markups. We construct cost and demand surprises from the simulated industry panel, calculate five-year trailing uncertainty using the same predetermined timing as in the data, and standardize the resulting measures. We then estimate the same fixed-effects regression with the same first-moment controls. The instrument is the one place where the empirical design cannot be reproduced exactly. Model industries are ex-ante identical and carry no input-output links to one another, so a leave-one-out shift-share of other industries' volatility would contain no relevant first-stage variation. We instead instrument measured uncertainty with the industry's own latent conditional volatility, the object the trailing window estimates with error. Because latent volatility is an exogenous state of the simulated model and is orthogonal to the sampling error in measured uncertainty, it provides the model counterpart to the measurement-error correction delivered by the empirical instrument.

The estimation uses four targeted moments for four structural parameters, so the model is exactly identified. Every parameter generally moves every moment, but the moments have natural primary identifying roles. Absolute risk aversion α primarily governs the magnitude of the two uncertainty coefficients. The demand slope b helps govern their relative magnitude through demand curvature. Idiosyncratic productivity dispersion $\sigma_{z,\text{idio}}$ primarily governs firm turnover by spreading firms around the exit threshold. The operating cost F primarily governs the aggregate markup by changing selection and the toughness of competition. We report the scaled numerical moment Jacobian and its singular values to assess whether the targeted moments provide independent local information about the four parameter directions.